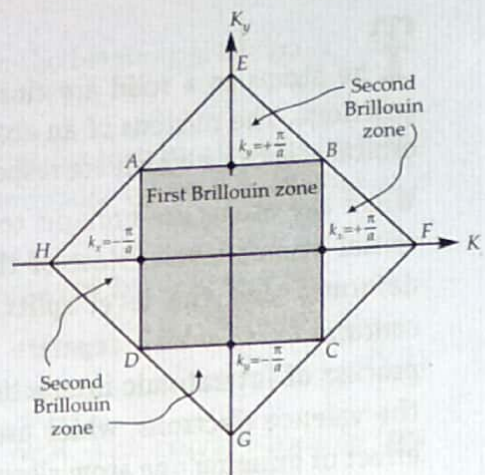


CHAPTER 9

Band Theory of Solids



"A useful way to visualize the difference between conductors, insulators and semiconductors is to plot the available energies for electrons in the materials."

– S.K. Kor

In This Chapter

- 9.1 Origin of Band Structure
- 9.2 Periodic Potential in Crystalline Solid
- 9.3 Wave Equation in a Periodic Potential and Bloch Theorem
- 9.4 The Kronig-Penney Model
- 9.5 Brillouin Zones
- 9.6 Variation of Energy and Velocity with Wave Vector
- 9.7 Variation of Effective mass with Wave Vector
- 9.8 Bohr's Atomic Model
- 9.9 Energy Levels
- 9.10 Energy Band
- 9.11 Classification of Solids on the basis of Energy bands
- 9.12 Semiconductors
- 9.13 Classification of Semiconductors
- 9.14 Intrinsic Semiconductors
- 9.15 Fermi Level : Effect of Temperature on Intrinsic Semiconductor
- 9.16 Fermi Level in Extrinsic Semiconductors
- 9.17 Variation of Fermi Level with Temperature in Extrinsic Semiconductors
- 9.18 The Hall Effect

The atoms in a solid are closely packed, but are in a state of vibration about their mean positions. The nucleus of an atom can be considered at rest due to its large mass and it is the orbital electrons which are responsible for most of the characteristic properties of the solids.

When the atoms are brought together in a solid, the interaction between them perturbs the initial atomic levels. Most of the symmetry of the electronic states in the isolated atom is deformed and the level splits into *bands* of levels. A *band constitutes a sort of energy continuum*, in which separate levels due to individual atom cannot be identified. In the process of interatomic interaction, the inner shell electron states are least affected, whereas the valence electrons, which are closest to the neighbouring ions, are the most affected. The effect of bringing one atom close to the other is to split sharp energy level into a pair of levels.

Let ψ_1 and ψ_2 be the electronic wave functions of the two atoms respectively when they are infinite distance apart. As the two atoms are brought together the wave functions ψ_1 and ψ_2 overlap. The resultant wave function is either $\psi_1 + \psi_2$ or $\psi_1 - \psi_2$ as shown in Figs. 9.1 (a) and (b).

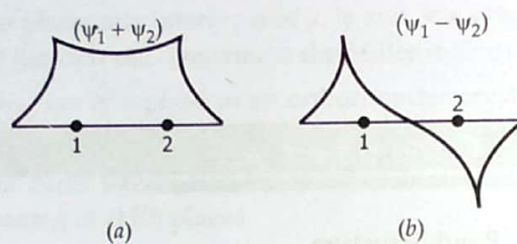


Fig. 9.1 Overlapping of ψ_1 and ψ_2

The electron has finite probability of existing midway between two nuclei for the wave functions $(\psi_1 + \psi_2)$. In this region the binding energy increases because the electron experiences the force of both the nuclei. For the wave function $(\psi_1 - \psi_2)$, the probability of finding electron between the nuclei is zero. Thus the extra contribution of energy disappears. In other words, there is a difference between the energy states $(\psi_1 + \psi_2)$ and $(\psi_1 - \psi_2)$. Thus when two isolated atoms are brought together each energy state splits up into two distinct energy states.

If there are many interacting atoms say N and N is very large, each energy state will split into N number of energy states. The separation between these energy states will be so small that each level will appear to have spread into a continuous, band of energy levels. This is why energy bands rather than specific energy levels exists in a solid. The width of a band depends upon the strength of interaction.

This explains how the atomic energy levels are split into bands when a number of atoms are brought close together to form a crystal.

In this chapter we will discuss Kronig-Penney model, Brillouin zone, effective mass of electron, important bands in solids and Hall effect on primary level.

9.1 ORIGIN OF BAND STRUCTURE

The band structure of a solid crystal can be described by the nearly free electron model in which the band electrons in a crystal are supposed to move in a region of periodically varying potential due to the ion cores of the crystal.

The energy of a free electron in terms of the wave vector $\mathbf{k}$ is given by

$$E_k = \frac{\hbar^2}{2m} k^2 = \frac{\hbar^2}{2m} (k_x^2 + k_y^2 + k_z^2) \quad \dots(9.1)$$

where k_x , k_y and k_z are respectively the x , y and z components of $\mathbf{k}$. The corresponding free electron wave function is of the form

$$\psi_k(\mathbf{r}) = e^{i(\mathbf{k} \cdot \mathbf{r})} = e^{i(k_x x + k_y y + k_z z)} \quad \dots(9.2)$$

where x , y , z are the components of position vector $\mathbf{r}$.

If we consider one dimensional case

$$\psi_k(x) = e^{ik_x x} \quad \text{or} \quad \psi = e^{ikx} \quad \dots(9.3)$$

[taking $\psi_k(x) = \psi$ and $k_x = k$]

It is also considered here that the interaction between electrons and ion cores in one dimensional lattice with ion spacing a as shown in Fig. 9.2.

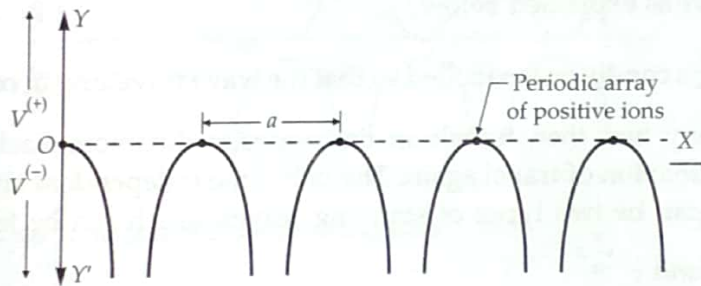


Fig. 9.2 Interaction between electrons and ions (1D)

The ion cores bear a positive charge and the electrostatic potential energy of an electron in the field of a positive ion is *attractive*, i.e., negative. This is why the potential energy is shown along X-axis (Fig. 9.2).

Low energy electrons can travel freely through a crystal since their wavelengths are long relative to the lattice spacing a . More energetic electrons like these having Fermi energy in a metal have wavelengths comparable with a . Such electrons suffer diffraction in the same way as X-rays or electrons in a beam directed at the crystal surface from outside. An electron having de Broglie wavelength λ undergoes Bragg reflection from one of the atomic planes in a crystal when it approaches the plane at an angle θ given by

$$(a + b) \sin \theta = n\lambda \quad \dots(9.4)$$

As lattice spacing $a = b$, $n\lambda = 2a \sin \theta$

In terms of wave vector $k = \frac{2\pi}{\lambda}$, the above relation becomes

$$k = \pm \frac{n\pi}{a \sin \theta} \quad \dots(9.5)$$

In a one-dimensional linear lattice $\theta = \frac{\pi}{2}$ and $k = \pm \frac{n\pi}{a}$. Thus a line representing k values is divided by energy discontinuities into segments of length $\pm \frac{\pi}{a}$. These line segments are known as Brillouin zones as shown in Fig. 9.3.

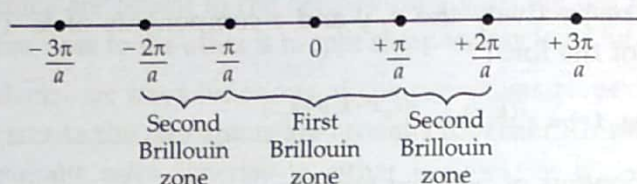


Fig. 9.3 Brillouin zones

A free electron travelling wave in one dimension, say along X-axis is given by

$$\psi = e^{ikx} \quad \dots(9.6)$$

At the boundaries $k = \pm \frac{\pi}{a}$, the wave functions are not travelling waves $e^{i\frac{\pi}{a}x}$ and $e^{-i\frac{\pi}{a}x}$, but these are standing waves as explained below :

For $k = \pm \frac{\pi}{a}$, Bragg's condition is satisfied so that the wave travelling in one direction is Bragg reflected at the boundary and then travels in the opposite direction. Each subsequent Bragg reflection reverses the direction of travel again. The only time independent situation is formed by standing waves. There can be two types of standing waves which can be formed from the two travelling waves $e^{+i\frac{\pi}{a}x}$ and $e^{-i\frac{\pi}{a}x}$.

$$\psi(+) = e^{+i\frac{\pi}{a}x} + e^{-i\frac{\pi}{a}x} = 2 \cos \frac{\pi}{a}x \quad \dots(9.7)$$

and

$$\psi(-) = e^{+i\frac{\pi}{a}x} - e^{-i\frac{\pi}{a}x} = 2i \sin \frac{\pi}{a}x \quad \dots(9.8)$$

The standing waves are labelled (+) if they do not change sign when $(-x)$ is substituted for $(+x)$ and (-) if they change sign.

1. The same result $k = \pm \frac{n\pi}{a}$ can be obtained from Bragg's reflection condition in reciprocal lattice which states

$$\left(k + \frac{G}{2}\right) \cdot G = 0$$

$$\text{or } k = \pm \frac{G}{2} = \pm \frac{\pi}{a} \quad \left[\because G = \pm \frac{2n\pi}{a} \right]$$

In quantum mechanics, the probability density of a particle $\rho = |\psi|^2$. For a pure travelling wave given by $\psi = e^{ikx}$, the probability density $\rho = e^{ikx} e^{-ikx} = 1$, so that the charge density $e|\psi|^2 = a$ constant, where e is the charge on the electron. However, the charge density is not constant for standing waves. For $\psi(+)$ charge density $e|\psi_+|^2 \propto \cos^2 \frac{\pi}{a} x$.

For $x = 0, a, 2a, \dots$ etc., $\cos^2 \frac{\pi}{a} x = 1$ and has a maximum value so that negative electron charge density is concentrated at the positive ion cores, thereby lowering the potential energy.

Similarly, for $\psi(-)$ charge density $e|\psi_-|^2 \propto \sin^2 \frac{\pi}{a} x$.

For $x = \frac{1}{2}a, \frac{3}{2}a, \dots$ etc.; $\sin^2 \frac{\pi}{a} x = 1$ and has a maximum value so that negative electron charge is concentrated midway between the positive ion cores, thereby raising the potential energy in comparison with that of the travelling wave. The distribution of electron charge density corresponding to a travelling wave and that for the standing wave $\psi(+)$ and $\psi(-)$ is shown in Fig. 9.4. If we calculate the average values of potential energy over the three distributions, we find that the potential energy $|\psi_+|^2$ is lower than that of a travelling wave, whereas the potential energy of $|\psi_-|^2$ is higher than that of a travelling wave.

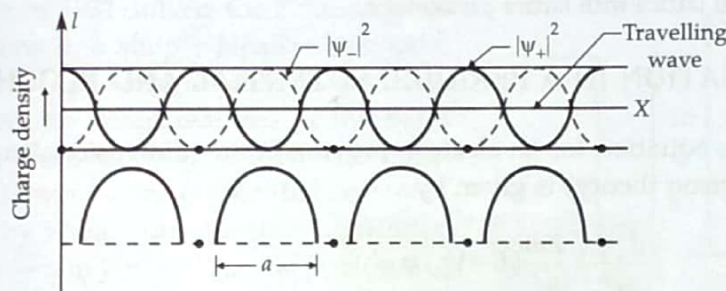


Fig. 9.4 Charge density corresponding to a travelling wave.

9.2 PERIODIC POTENTIAL IN CRYSTALLINE SOLIDS

In solids one deals with a large number of interacting particles and hence the problem of determining the electron wave functions and the energy levels is extremely complicated. A simple quantum mechanical picture of an electron in a crystal can, however, be obtained by assuming that the atomic nuclei are at rest in the crystalline state and that the electron is in a periodic potential which has periodicity of the lattice. The periodic potential may be considered to be caused by the fixed nuclei plus some average potential due to all other electrons. The solutions of Schrödinger equation for this potential gives a set of state or levels which may be occupied by the electrons according to Pauli's exclusive principle.

The potential energy of an electron at a distance r from an atomic nucleus charge Ze is given by

$$U = \frac{-Ze^2}{4\pi\epsilon_0 r} \quad \dots(9.9)$$

The variation of potential energy V with distance r for an isolated system is shown in Fig. 9.5.

To form a crystal, a number of such nuclei are brought close each other. Therefore, the potential energy of an electron is the sum of potential energies due to the individual nuclei. Hence, the variation of potential energy with distance for a one-dimensional crystal is as shown in Fig. 9.6.

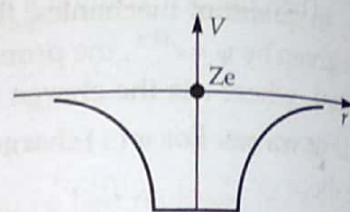


Fig. 9.5 Variation of V with r .

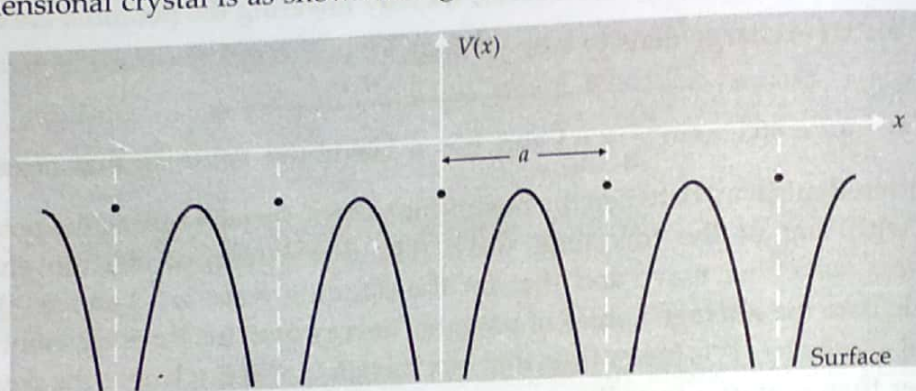


Fig. 9.6 One-dimensional representation of potential experienced by an electron in a perfectly periodic crystal lattice with lattice parameter a .

9.3 WAVE EQUATION IN A PERIODIC POTENTIAL AND BLOCH THEOREM

The Schrödinger wave equation for an electron moving in one dimensional constant potential V_0 (according to free electron theory) is given by

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \quad \dots(9.10)$$

The solution of this equation is given by

$$\psi(x) = e^{\pm ikx} \quad \dots(9.11)$$

where k represents the momentum of the electron (p) divided by $\hbar$. From above equation, we get kinetic energy of the electron

$$K = E - V_0 = \frac{\hbar^2 k^2}{2m} = \frac{p^2}{2m} \quad \dots(9.12)$$

The complete solution for the wave function containing the time is obtained by multiplying $\psi(x)$ by $e^{i\omega t}$

where

$$\omega = \frac{E}{\hbar} \quad \dots(9.13)$$

Now consider the Schrödinger equation for an electron moving in one dimensional potential. Thus, let the potential energy of electron satisfy the equation

$$U(x) = U(x + a) \quad \dots(9.14)$$

where a is the period.

Then the Schrödinger's equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)]\psi = 0 \quad \dots(9.15)$$

With reference to the solutions of this equation, there is an important theorem which states that there exist solutions of the form

$$\begin{aligned} U_k(x) &= e^{\pm ikx} U_k(x) \\ U_k(x+a) &= e^{\pm ika} U_k(x) \end{aligned} \quad \dots(9.16)$$

where $U_k(x) = U_k(x+a)$

i.e., the solutions are plane wave modulated by the function $U_k(x)$ which has the same periodicity as the lattice. This theorem is known as *Bloch theorem*. Thus Bloch theorem is a mathematical statement regarding the form of the one electron wave functions for a perfectly periodic potential.

Expression (9.16) is known as *Bloch function*.

9.4 THE KRONIG-PENNEY MODEL (K-P MODEL)

The Kronig-Penney model (named after Ralph Kronig and William Penney) is a simple, idealized quantum mechanical system that consists of an infinite array of rectangular potential barriers. Kronig and Penney in 1931 solved the Schrödinger's equation for electrons in a simple idealized periodic field. The model is of considerable importance because it interprets the main features of the band structure of metals. The periodic field was assumed in the form of potential wells of zero potential energy of width a separated by rectangular barriers of width b and height V_0 as shown in Fig. 9.7. For the potential of Fig. 9.7, the Schrödinger equation has to be solved in two regions :

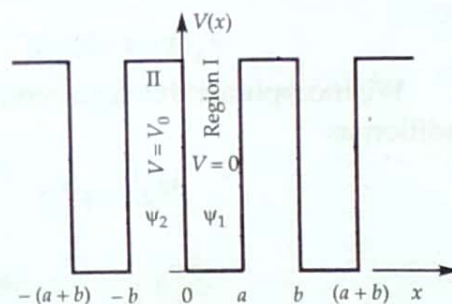


Fig. 9.7 One dimensional chain of potential wells.

- (a) **Region I**, $0 < x < a$, potential well region in which $V = 0$ and
- (b) **Region II**, $-b < x < 0$, barrier region in which $V = V_0$.

In order to find the allowed energies, we solve the Schrödinger's equation in these two regions and apply the appropriate boundary condition.

In region I, the Schrödinger's equation is

$$\frac{d^2\psi}{dx^2} + \frac{2mE}{\hbar^2} \psi = 0$$

or

$$\frac{d^2\psi_1}{dx^2} + \beta^2 \psi_1 = 0 \quad \left[\text{Here, } \beta^2 = \frac{2mE}{\hbar^2} \right] \quad \dots(9.17)$$

Solution of this equation is

$$\psi_1 = Ae^{i\beta x} + Be^{-i\beta x}, \quad \text{where } A \text{ and } B \text{ are constants.} \quad \dots(9.18)$$

In region II, the Schrödinger's equation is

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0$$

or
$$\frac{d^2\psi_2}{dx^2} - \alpha^2\psi_2 = 0 \quad \left[\text{Here, } \alpha^2 = \frac{2m(E - V_0)}{\hbar^2} \right] \quad \dots(9.19)$$

Solution of this equation, for $E < V_0$ is

$$\psi_2 = Ce^{\alpha x} + De^{-\alpha x} \quad \dots(9.20)$$

where C and D are constants.

For continuity at $x=0$

$$\psi_1(0) = \psi_2(0) \quad \dots(9.21)$$

$$\left. \frac{\partial \psi_1}{\partial x} \right|_{x=0} = \left. \frac{\partial \psi_2}{\partial x} \right|_{x=0} \quad \dots(9.22)$$

In a periodic lattice $U(x+a) = U(x)$, it is expected that the wavefunction will also exhibit this periodicity. Since the wave function must be a Bloch wave function, therefore Eq. (9.17) gives

$$\psi_k(x+a+b) = e^{-ik(a+b)}\psi_k(x) \quad \dots(9.23)$$

We incorporate this requirement at $x=-b$ and $x=a$, then we obtain the continuity boundary condition as

$$\psi_2|_{x=-b} = e^{-ik(a+b)}\psi_1|_{x=a} \quad \dots(9.24)$$

$$\left. \frac{\partial \psi_2}{\partial x} \right|_{x=-b} = e^{-ik(a+b)} \left. \frac{\partial \psi_1}{\partial x} \right|_{x=a} \quad \dots(9.25)$$

When we apply the boundary conditions, Eqs. (9.21), (9.22), (9.23) and (9.24), we obtain four equations for four unknowns

$$A + B = C + D \quad \dots(9.26)$$

$$i\beta(A - B) = \alpha(C - D) \quad \dots(9.27)$$

$$Ce^{-\alpha b} + De^{\alpha b} = e^{-ik(a+b)}[Ae^{i\beta a} + Be^{-i\beta a}] \quad \dots(9.28)$$

$$\alpha Ce^{-\alpha b} - \alpha De^{\alpha b} = e^{-ik(a+b)} i\beta[Ae^{i\beta a} - Be^{-i\beta a}] \quad \dots(9.29)$$

By solving these equations simultaneously e.g., by requiring that the determinants of the coefficients of A, B, C and D vanish, we get

$$\begin{vmatrix} 1 & 1 & -1 & -1 \\ i\beta & -i\beta & -\alpha & \alpha \\ -e^{-ik(a+b)+i\beta a} & -e^{-ik(a+b)-i\beta a} & e^{-\alpha b} & e^{\alpha b} \\ i\beta e^{-ik(a+b)+i\beta a} & -i\beta e^{-ik(a+b)-i\beta a} & -\alpha e^{-\alpha b} & \alpha e^{\alpha b} \end{vmatrix} = 0 \quad \dots(9.30)$$

On solving the determinant Eq. (9.30) and after simplification, we get

$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} \sinh \alpha b \sin \beta a + \cosh \alpha b \cos \beta a = \cos(a+b)k \quad \dots(9.31)$$

To simplify Eq. (9.31), Kronig and Penney considered the case when $V_0 \rightarrow \infty$ and $b \rightarrow 0$, but the product $V_0 b$ has a finite value *i.e.*, the potential barriers become *delta functions*. Under these conditions, the model is modified in such a way that represent a series of well separated by infinitely thin potential barriers of infinitely large potential. The limiting value of $V_0 b$ for $V_0 \rightarrow \infty$ and $b \rightarrow 0$ is known as *barrier strength*.

As $b \rightarrow 0$, $\sinh \alpha b \rightarrow \alpha b$ and $\cosh \alpha b \rightarrow 1$

Also from Eqs. (9.17) and (9.19)

$$\alpha^2 = \frac{2m(E - V_0)}{\hbar^2} \quad \text{and} \quad \beta^2 = \frac{2mE}{\hbar^2}$$

Then
$$\beta^2 - \alpha^2 = \frac{2mV_0}{\hbar^2}$$

or
$$\frac{\beta^2 - \alpha^2}{2\alpha\beta} = \frac{mV_0}{\alpha\beta\hbar^2} \quad \dots(9.32)$$

Putting the value $\frac{\beta^2 - \alpha^2}{2\alpha\beta}$ from Eq. (9.32) in Eq. (9.31), we get

$$\begin{aligned} \frac{mV_0}{\alpha\beta\hbar^2} \alpha b \sin \beta a + \cos \beta a &= \cos ka \\ \frac{mV_0 b}{\beta\hbar^2} \sin \beta a + \cos \beta a &= \cos ka \end{aligned} \quad \dots(9.33)$$

Let us define a quantity $p = \frac{mV_0 b a}{\hbar^2}$, which is measure of the area $V_0 b$ of the potential barrier, then

$$p \frac{\sin \beta a}{\beta a} + \cos \beta a = \cos ka \quad \dots(9.34)$$

The physical significance of the quantity p is that if p is increased and the given vector is bound more strongly to a particular potential well.

When $p \rightarrow 0$, then potential barrier becomes very weak which means that electrons are free electrons. In this, we obtain from Eq. (9.34),

$$\beta a = ka \quad \text{or} \quad \beta = k$$

320

Now

$$\beta^2 = \frac{2mE}{\hbar^2} = k^2$$

Hence

$$E = \frac{k^2 \hbar^2}{2m} = \frac{\hbar^2 k^2}{8\pi^2 m}$$

This result is similar as obtained by free electron theory. Equation (9.34) also gives a condition, which must be satisfied so that solutions of the wave equation may exist. Since βa can have values between $+1$ and -1 . Such values of βa , therefore, represent wave like solutions of the form

$$U(x) = e^{ikx} U_k(x)$$

and are *allowed values*. The other values of βa are not allowed.

If we plot a graph between $\left(\frac{p \sin \beta a}{\beta a} + \cos \beta a \right)$ and βa for the value of $p = \frac{3a}{2}$, we get the graph as shown in Fig. 9.8.

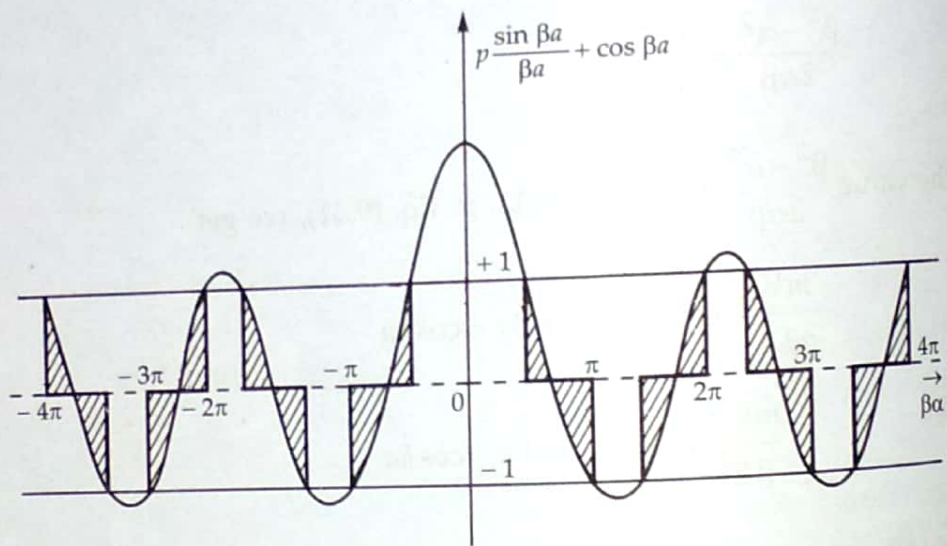


Fig. 9.8 Plot of $\frac{p}{\beta a} \sin \beta a + \cos \beta a$ vs. βa to show allowed and forbidden band for the K.P. approximation.

Conclusions of Kronig-Penney Model

On the basis of above analysis, we conclude following interesting points :

(i) Figure 9.8 depicts that energy spectrum consists of an infinite number of *allowed regions* (shown **thick**) separated by intervals in which there are no energy levels (shown **dotted**) *forbidden regions* or *bands*. The allowed energy level boundaries correspond to the value of $\cos \beta a$

$$\text{or } ka = n\pi \quad \text{or } k = \frac{n\pi}{a}$$

(ii) As the first term $p \frac{\sin \beta a}{\beta a}$ of Eq. (9.34) decreases with increasing βa , the width of the allowed bands increases with increasing the value of βa . Hence, forbidden energy region becomes narrow. This is seen in Fig. 9.9, where energy bands have increasing width.

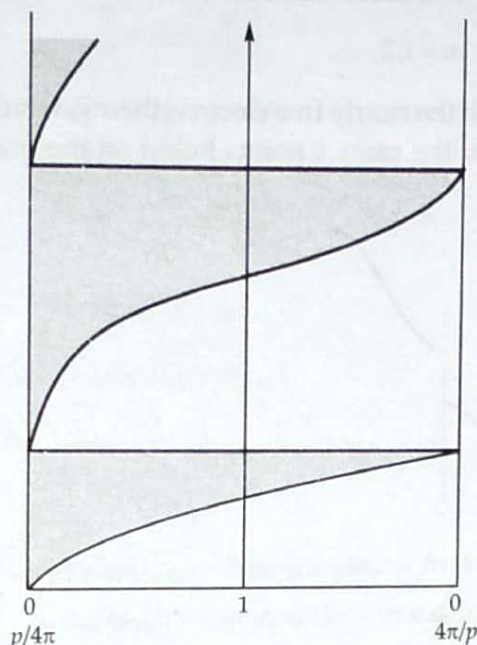


Fig. 9.9 The width of allowed bands as a function of p . The allowed regions are shaded.

(iii) The width of the particular allowed band decreases with increasing p , that is with increasing binding energy of the electrons. $p=0$ corresponds to free electrons.

(iv) In the extreme case where p tends to ∞ , the allowed regions become infinitely narrow and the energy spectrum becomes a line spectrum. In this case, Eq. (9.34) will have solutions if

$$\sin \beta a = 0, \text{ then } \beta a = \pm n\pi \text{ with } n = 1, 2, 3, \dots$$

Therefore, the energy spectrum is given by

$$\beta^2 = \frac{2mE}{\hbar^2}$$

$$E = \frac{\left(\frac{n\pi}{a}\right)^2 \hbar^2}{2m} \quad \dots(9.37)$$

As $\beta = \pm \frac{n\pi}{a}$;

$$E_n = \frac{\pi^2 \hbar^2}{2ma^2} n^2 \quad \text{or } p \rightarrow \infty$$

which is the energy levels of a particle in a box of atomic dimension because for large p , tunneling through the barriers becomes impossible.

(v) From Eq. (9.34), it is possible to obtain the energy E as a function of the wave vector k

$$k = \pm \frac{n\pi}{a}$$

Equation (9.37) can be written as

$$E = \frac{k^2 \hbar^2}{2m} \quad \dots(9.38)$$

This is shown in Fig. 9.10. The discontinuities in $E-k$ curve occur for

$$k = \pm \frac{n\pi}{a}; \quad n=1,2,\dots \quad \dots(9.39)$$

This inference is in line with the nearly free electron theory, which also brought at the possible discontinuities in the $E-k$ curve at the same k points based on the Bragg reflection of electrons.

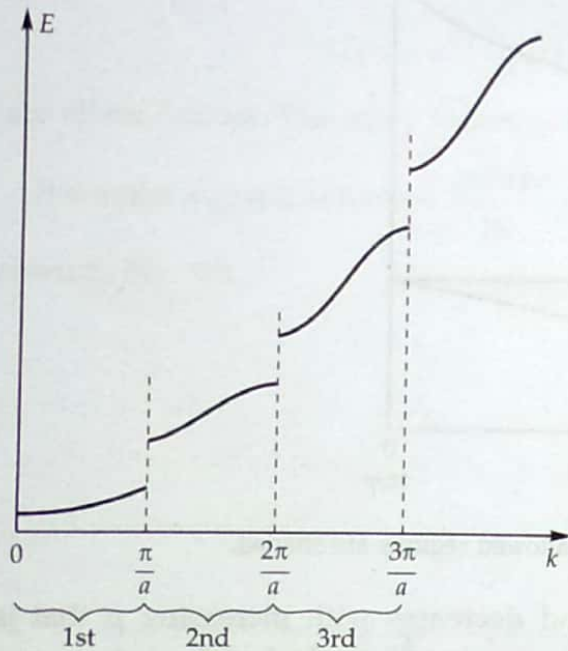


Fig. 9.10 $E-k$ curve—different positive Brillouin zone boundaries are marked.

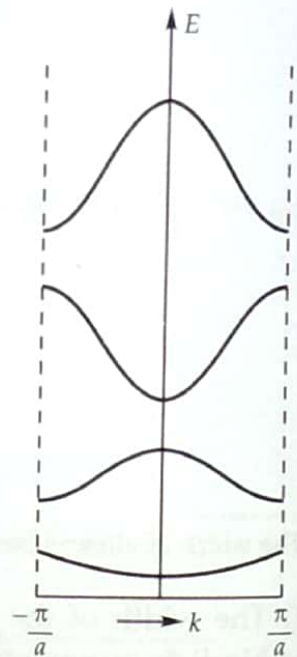


Fig. 9.11 Energy at reduced k points (in the first zone)

(vi) Within a given energy band, the energy is a periodic function of k . For example, if k is replaced by $k + \frac{2n\pi}{a}$, n is an integer; the R.H.S of Eq. (9.34) remains the same. Thus, it is convenient to introduce the reduced wave vector which is limited in the region $-\frac{\pi}{a} \leq k \leq +\frac{\pi}{a}$. The energy in the reduced wave vector is represented in Fig. 9.11.

The fact that k is not uniquely determined, can also be realized from Bloch function.

Example 9.1 In Kronig-Penney model, determine the energy of the lowest band for $p \ll 1$.

Solution. Equation, $\cos ka = \frac{p}{a\beta} \sin \beta a + \cos \beta a$

gives the energy of lowest band corresponding to $k = \pm \frac{\pi}{a}$ as $\frac{p}{a\beta} \sin \beta a + \cos \beta a = \pm 1$

Consider only the magnitude on R.H.S., then

$$\frac{p}{a\beta} \cdot 2 \sin \frac{\beta a}{2} \cos \frac{\beta a}{2} = 1 - \cos \beta a = 2 \sin^2 \frac{\beta a}{2}$$

$$\text{or} \quad \frac{p}{a\beta} \cdot 2 \sin \frac{\beta a}{2} \cos \frac{\beta a}{2} = 1 - \cos \beta a = 2 \sin^2 \frac{\beta a}{2}$$

$$\text{or} \quad \tan \frac{\beta a}{2} = \frac{p}{a\beta}$$

$$\text{If } p \ll 1, \text{ then} \quad \frac{p}{a\beta} = \tan \left(\frac{p}{a\beta} \right)$$

$$\text{Here} \quad \frac{\beta a}{2} = \frac{p}{a\beta}$$

$$\text{or} \quad \beta^2 = \frac{2p}{a^2} = \frac{2mE}{\hbar^2}$$

$$\text{or} \quad E = \frac{\hbar^2}{ma^2} p$$

9.5 BRILLOUIN ZONES

9.5.1 Linear Lattice

According to Kronig and Penney, the discontinuities in a one dimensional lattice occur when $k = \frac{n\pi}{a}$, where n is a positive or negative integer. In a one dimensional mono-atomic lattice a line representing the value of k is divided by energy discontinuities into segment of length as shown in Fig. 9.3. These line segments are known as Brillouin Zones.

The first reflection and the first energy gap occurs at $k = \pm \frac{\pi}{a}$. The reflection at $k = \pm \frac{\pi}{a}$ arises because the wave reflected from one atom in a linear lattice interferes constructively with the wave from a nearest neighbour atom, the phase difference between the two reflected waves for the values of $k = +\frac{\pi}{a}$ and $k = -\frac{\pi}{a}$ being $\frac{2\pi}{a}$. The region in k -space between $-\frac{\pi}{a}$ and $+\frac{\pi}{a}$, i.e., the segment $-\frac{\pi}{a} \leq k \leq +\frac{\pi}{a}$ is called the first Brillouin Zone.

The second Brillouin zone contains electrons with $k > \frac{\pi}{a}$, i.e., electrons having k -values between $\frac{\pi}{a}$ and $\frac{2\pi}{a}$ for electrons moving in the $\pm x$ direction. The second reflection and second energy gap occurs at $k = \pm \frac{2\pi}{a}$ and therefore, the region in the k space between $-\frac{2\pi}{a}$ to $-\frac{\pi}{a}$ and $+\frac{\pi}{a}$ to $+\frac{2\pi}{a}$, i.e., the segment $-\frac{2\pi}{a} \leq k \leq -\frac{\pi}{a}$ and $+\frac{\pi}{a} \leq k \leq \frac{2\pi}{a}$ forms the second Brillouin zone of the lattice. Similarly, we can discuss the formation of third Brillouin zone ; ... etc.

The first and second Brillouin zones and the energy gaps are shown in Fig. 9.12 by plotting value of E in eV corresponding to various values of k .

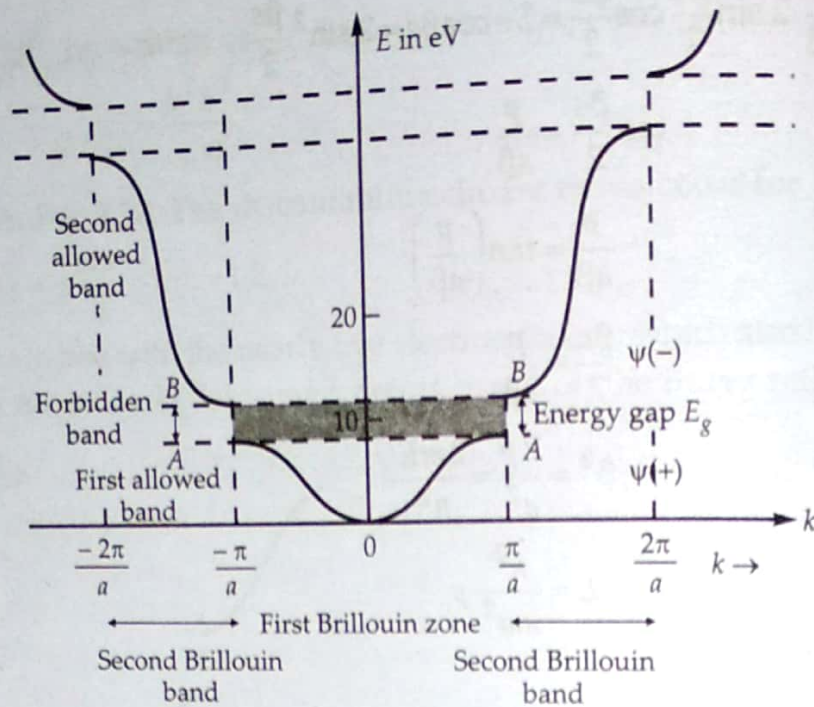


Fig. 9.12 The first and second BZ and energy gap

9.5.2 Two Dimensional Lattice

Proceeding as in the case of a linear lattice, the first zone for a two dimensional lattice in the X-Y plane is the square ABCD the boundaries of which are given by the relation $k_x = +\frac{\pi}{a}$ and $k_x = -\frac{\pi}{a}$ and $k_y = +\frac{\pi}{a}$ and $k_y = -\frac{\pi}{a}$ as shown in Fig. 9.13.

The second Brillouin zone for a two dimensional lattice in the X-Y plane is the square EFGH the boundary of which are given by the relation

$$k_x = \pm \frac{2\pi}{a}; \quad k_y = \pm \frac{2\pi}{a}$$

...(9.40)

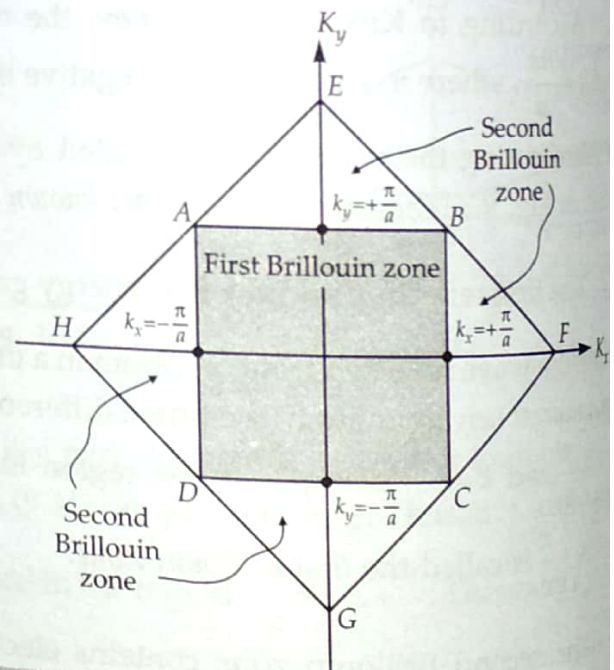


Fig. 9.13 First and second BZ in 2D lattice

9.6 VARIATION OF ENERGY AND VELOCITY WITH WAVE VECTOR

According to de-Broglie, an electron moving with a velocity is equivalent to a wave packet moving with a group velocity v_g .

The particle velocity is equal to the group velocity and is given by

$$v = v_g = \frac{d\omega}{dk}$$

...(9.41)

where ω is the angular frequency ($2\pi\nu$) of $|k|$ the wave vector ($2\pi/\lambda$). Also the energy of particle in terms of associated wave is given by

$$E = \hbar\omega \quad \text{or} \quad \omega = \frac{E}{\hbar}$$

$$\therefore v_g = v = \frac{d\omega}{dk} = \frac{1}{\hbar} \frac{dE}{dk} \quad \dots(9.42)$$

For a free electron $E = \frac{\hbar^2 k^2}{2m} \quad \dots(9.43)$

or $\frac{dE}{dk} = \frac{\hbar^2}{m} k$

$$\therefore v = \frac{1}{\hbar} \frac{dE}{dk} = \frac{1}{\hbar} \frac{\hbar^2}{m} k = \frac{p}{m} \quad \dots(9.44)$$

where p is the momentum of the particle.

9.6.1 Variation of E with k

On the band theory E is not in general proportional to k^2 as given in Eq. (9.43) as shown in Fig. 9.14(a). The curve is symmetrical about E -axis having point of inflexion at M and N where $\frac{dE}{dk}$ is maximum. As $v = \frac{1}{\hbar} \frac{dE}{dk}$; v is maximum at value of k corresponding to M and N . At the points A , O and B , the slope $\frac{dE}{dk} = 0$. The value of the slope is different for different portion of the curve.

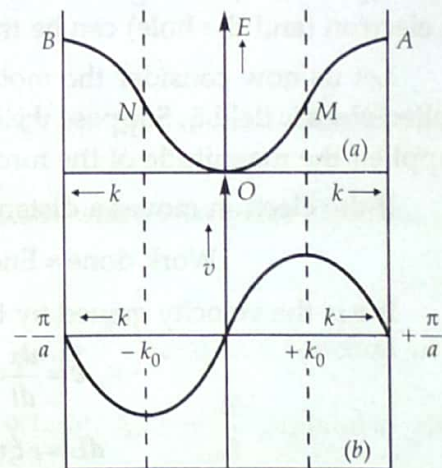


Fig. 9.14 Energy and velocity variation with k .

9.6.2 Variation of v and k

The variation of v with k is shown in Fig. 9.12(b). At $k = 0$, $k = +\frac{\pi}{a}$ and $k = -\frac{\pi}{a}$, the velocity of the electron is zero. Thus the velocity of the electron is zero at the bottom as well as at the top of the Brillouin zone. At intermediate regions in zone the velocity is maximum at the points of inflexion $k = k_0$ and reaches the free electron velocity. Beyond this point the velocity decreases with increasing energy which is different in behaviour from that of free electron.

Example 9.2 Find the ratio between the kinetic energies of an electron in a two dimensional square lattice (a) when $k_x = k_y = \pi/a$ and (b) when $k_x = \pi/a$ and $k_y = 0$.

Solution. Given : (a) $k_x = k_y = \pi/a$ (b) $k_x = \pi/a$ and $k_y = 0$.

We know that the energy of an electron inside a Brillouin zone is given by

$$E = \frac{\hbar^2}{8\pi^2 m} k^2$$

For case I, the energy becomes

$$E_1 = \frac{h^2}{8\pi^2 m} (k_x^2 + k_y^2) = \frac{h^2}{8\pi^2 m} \times \frac{2\pi^2}{a^2} = \frac{h^2}{4a^2 m}$$

Similarly, for case II, $E_2 = \frac{h^2}{8\pi^2 m} \frac{\pi^2}{a^2} = \frac{h^2}{8a^2 m}$

This gives, $\frac{E_1}{E_2} = 2$

9.7 EFFECTIVE MASS VARIATION AS A FUNCTION OF WAVE VECTOR

The electrons in a crystal are not completely free but *interact with the periodic potential* of the crystal lattice. As a result their wave particle motion is different from that of an electron in free space. The while using the equations of electrodynamics for charge carriers in a crystal, we must use the value of particle mass or *effective mass* after taking into account the effect of the crystal lattice so that the electron (and the hole) can be treated as almost free carriers of charge for the calculations.

Let us now consider the motion of an electron in a crystal in the presence of an external applied electric field ξ . Suppose the electron is initially in a state k , when the external electric field is applied the magnitude of the force acting on the electron is $e\xi$.

If the electron moves a distance dx in a small time dt , under the action of this force, then

$$\text{Work done} = \text{Energy gained } dE = e\xi dx$$

If v is the velocity gained by the electron, then

$$v = \frac{dx}{dt} \text{ or } dx = v dt$$

$$\therefore dE = e\xi v dt$$

Now $v = \frac{1}{\hbar} \frac{dE}{dk}$

$$\therefore dE = \frac{e\xi}{\hbar} \frac{dE}{dk} dt$$

or $\frac{dk}{dt} = \frac{e\xi}{\hbar}$

Now $\hbar k = p$ and $\hbar \frac{dk}{dt} = \frac{dp}{dt} = F$

$$\therefore \hbar \frac{dk}{dt} = e\xi = F$$

Differentiating Eq. (9.47) with respect to t , we get

$$\frac{dv}{dt} = \frac{1}{\hbar} \frac{d^2 E}{dk dt} = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \frac{dk}{dt}$$

From Eq. (9.50), $\frac{dk}{dt} = \frac{F}{\hbar}$

Also $\frac{dv}{dt} = a$ (the acceleration)

From Eq. (9.51), $a = \frac{1}{\hbar} \frac{d^2 E}{dk^2} \cdot \frac{F}{\hbar}$

or $\frac{a}{F} = \frac{1}{\hbar^2} \frac{d^2 E}{dk^2}$... (9.52)

Taking $m^* = \frac{F}{a}$ as the effective mass of electron in the crystal, we have

$$m^* = \frac{\hbar^2}{d^2 E / dk^2} \quad \dots (9.53)$$

In some cases, a factor f_k is also used. It gives the ratio of the rest mass of a free electron m to its effective mass in the crystal in the k -state

$$f_k = \frac{m}{m^*} = \frac{m}{\hbar^2} \left(\frac{d^2 E}{dk^2} \right) \quad \dots (9.54)$$

This expression determines the extent to which the electron in the k -state behaves as a free electron.

If $f_k = 1$, $m^* = m$, the electron in the crystal in the k -state behaves as a free electron.

Variation of Effective Mass with k

Equation (9.53) shows the effective mass depends upon $\frac{d^2 E}{dk^2}$ and is a function of k . The variation of electron velocity v with k is shown in Fig. 9.14(b). As $v \propto \frac{dE}{dk}$, this also gives the variation of $\frac{dE}{dk}$ with k .

Now $\frac{d^2 E}{dk^2}$ is the slope of $\frac{dE}{dk}$ versus k curves and the value of $\frac{d^2 E}{dk^2}$ is positive for values of $k = -k_0$ to $k = +k_0$ which represents the lower half of the energy band. $\frac{d^2 E}{dk^2}$ is negative for values of $k = +k_0$ to $+\frac{\pi}{a}$ and $k = -k_0$ to $-\frac{\pi}{a}$, which represents the upper half of the energy band and m^* is negative at top i.e., upper half of the energy band. At the point of inflexion of E - k curve m^* becomes infinite. The variation of mass m^* with k from $k = -\frac{\pi}{a}$ to $k = +\frac{\pi}{a}$ is shown in Fig. 9.15.

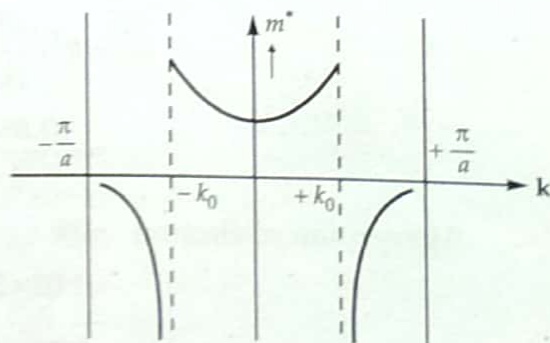


Fig. 9.15 Variation of effective mass with k .

The large effective mass can be interpreted as being due to strong binding force between the electron and the lattice for some value of k . The negative mass may be interpreted in terms of Bragg reflection when k is close to $\pm \frac{\pi}{a}$, $\pm \frac{2\pi}{a}$, on account of which a force in one direction leads to a gain in momentum in opposite direction, which results in negative effective mass. The results of free electron theory can be applied in the band theory by simply replacing m by m^* . Thus Fermi energy in a metal is given by

$$E_F = \frac{\hbar^2}{2m^*} \left[3\pi^2 \frac{N}{V} \right]^{2/3} = \frac{\hbar^2}{2m^*} \left[\frac{3N}{8\pi V} \right] \quad \text{---(9.5)}$$

Physical Significance of Effective Mass

In most conductors $m^* = m$ because the valence band is only partially filled. But in semiconductors and insulators having almost filled valence bands the effective mass plays an important role in conduction process and provides a satisfactory description of the charge carriers in crystals. In crystals, such as alkali metals which have partially filled energy band, the conduction takes place mainly through electrons. But in crystals which have energy bands nearly full except for a few electron vacancies near the tip of the band, these negative charge and negative mass vacancies may be considered as positive charge and positive mass particle called *holes* which act as positive charge carriers to bring out conduction.

The concept of effective mass is able to account for many observed effects like 'electronic specific heat of transition metals' and their high paramagnetic susceptibility.

Example 9.3 The energy near the valence band edge of a crystal is given by

$$E = -10^{-39} k^2 \text{ Jm}^2$$

An electron with wave vector $10^{10} \hat{k}_x \text{ m}^{-1}$ is removed from an orbital in the completely filled valence band. Determine effective mass and momentum.

Solution. Given $E = -10^{-39} k^2$

Hence
$$\frac{d^2 E}{dk^2} = -2 \times 10^{-39}$$

Effective mass of the electron

$$\begin{aligned} m^* &= \hbar^2 / \frac{\partial^2 E}{\partial k^2} \\ &= - \frac{(1.05 \times 10^{-34})^2}{2 \times 10^{-39}} = -5.5 \times 10^{-30} \text{ kg} \end{aligned}$$

Momentum of electron $= \hbar k$

$$\begin{aligned} &= 1.05 \times 10^{-34} \times 10^{10} \hat{k}_x \\ &= 1.05 \times 10^{-24} \hat{k}_x \text{ Js m}^{-1}. \end{aligned}$$

9.8 BOHR'S ATOMIC MODEL

In 1913, Neil Bohr, Danish physicist gave clear explanation of atomic structure. According to Bohr :

- (i) An atom consists of a positively charged nucleus around which negatively charged electrons revolve in different circular orbits.
- (ii) The electrons can revolve round the nucleus only in certain permitted orbits *i.e.*, orbits of certain radii are allowed.
- (iii) The electrons in each permitted orbit have a certain fixed amount of energy. The larger the orbit (*i.e.*, larger radius) the greater is the energy of electrons.
- (iv) If electron is given additional energy (*e.g.*, heat, light etc.) it is lifted to higher orbit. The atom is said to be in state of excitation. This state does not last long, because electron soon falls back to the original lower orbit. As it falls, it gives back the acquired energy in the form of heat, light or other radiation.

Figure 9.16 shows the structure of silicon atom. It has 14 electrons.

Two electrons revolve in the first orbit, 8 in the second orbit and 4 in the third orbit. The first, second, third orbits etc. are also known as K, L, M orbits respectively.

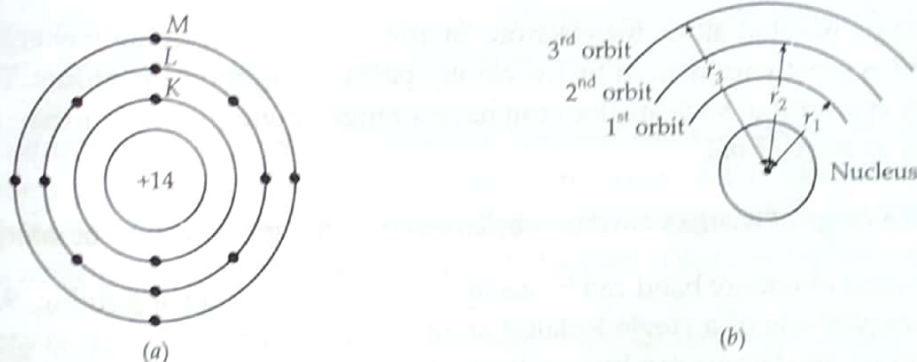


Fig. 9.16 Structure of silicon atom.

These electrons can revolve only in permitted orbits (*i.e.*, orbits or radii² r_1 , r_2 and r_3) and not in any arbitrary orbit. Thus, all radii between r_1 and r_2 or between r_2 and r_3 are forbidden. Each orbit has fixed amount of energy associated with it. If an electron in the first orbit is to be lifted to the second orbit, just the³ right amount of energy should be supplied to it. When this electron jumps from the second orbit to first, it will give back the acquired energy in the form of electromagnetic radiations.

9.9 ENERGY LEVELS

We know that each orbit has fixed amount of energy associated with it. The electrons moving in a particular orbit possess the energy of that orbit. The larger the orbit, the greater is its energy. It becomes clear that outer orbit electrons possess more energy than the inner orbit electrons.

2. The values of radii are determined from quantum consideration.
3. So that its total energy is equal to that of second orbit.

A convenient way of representing the energy of different orbits is shown in Fig. 9.18(ii). The first orbit represents *first energy level*, the second orbit indicates the *second energy level* and so on. The larger the orbit of an electron, the greater is energy and higher is the energy level.

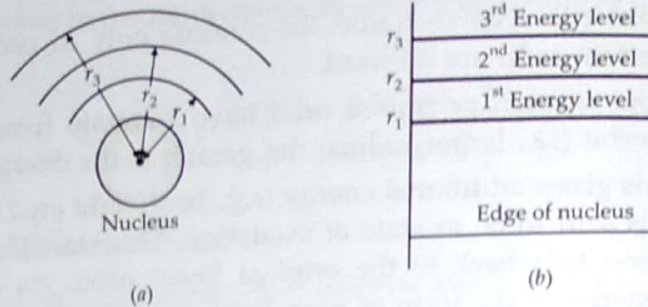


Fig. 9.17 Energy levels

9.10 ENERGY BAND

In case of a single isolated atom, the electrons in any orbit possess definite energy. However, an atom in a solid is greatly influenced by the closely-packed neighbouring atoms. The result is that the electron in any orbit of such an atom can have a range of energies rather than a single energy. This is known as *energy band*.

"The range of energies possessed by an electron in a solid is known as energy band."

The concept of energy band can be easily understood by referring to Fig. 9.18. Fig. 9.18(iii) shows the energy levels of a single isolated atom of silicon. Each orbit of an atom has a single energy. Therefore, an electron can have only single energy corresponding to the orbit in which it exists. However, when the atom is in a solid, the electron in any orbit can have a range of energies. For instance, electrons in the first orbit have slightly different energies because no two electrons in this orbit see exactly the same charge environment. Since there are millions of first orbit electrons, the slightly different energy levels form a band, called *first energy band* [Fig. 9.18(iii)]. The electrons in the first orbit can have any energy range in this band. Similarly, second orbit electrons form *second energy band* and so on.

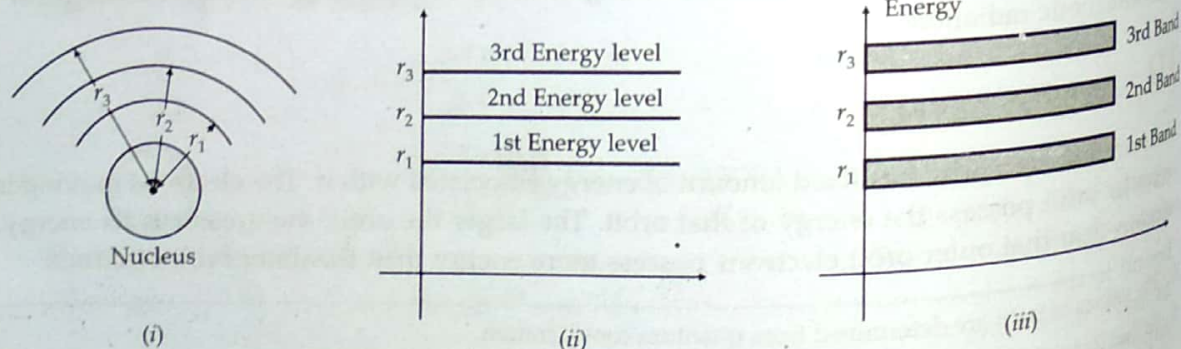


Fig. 9.18 Energy levels with energy bands.

Important Energy Bands in Solids

We know that, individuals $K, L, M, \dots$ energy levels of an isolated atom are converted into corresponding bands, when the atom is in a solid. Though there are a number of energy bands in solids, the following are of the particular importance [Fig. 9.19] :

(i) **Valence band.** The electrons in outermost orbit of an atom are known as valence electrons. These electrons have highest energy. In case of solid, valence electrons are confined in a band or energy range.

The range of energy (i.e., band) possessed by valence electrons is known as **valence band**.

This band may be completely or partially filled. In short we can say that *the upper-most energy band of solid which is partially or completely filled by the electrons, is called as valence band*. For instance, in case of inert gases, the valence band is full whereas for other materials, it is only partially filled. The partially filled band can accommodate more electrons.

(ii) **Conduction band.** In certain materials (e.g., metals), the valence electrons are loosely attached to the nucleus. Even at ordinary temperature, some of the valence electrons may get detached to become free electrons. In fact, it is these free electrons which are responsible for the conduction of current in a conductor. For this reason, they are called conduction electrons.

The range of energy (i.e., band) possessed by conduction electrons is known as **conduction band**.

All electrons in the conduction band are free electrons. If a substance has empty conduction band, it means that current conduction is not possible in that substance. Generally, insulators have empty conduction band. On the other hand, it is partially filled for conductors.

(iii) **Forbidden energy gap.**

The separation between conduction band and valence band on the energy level diagram is known as **forbidden energy gap**.

No electron of a solid can stay in a forbidden energy gap as there is no allowed energy state in this region. The width of the forbidden energy gap is measure of the boundage of valence electrons to the atom. The greater the energy gap, more tightly the valence electrons are bound to the nucleus. In order to push an electron from valence band to conduction band (i.e., to make the valence electron free), external energy equal to forbidden energy gap must be supplied.

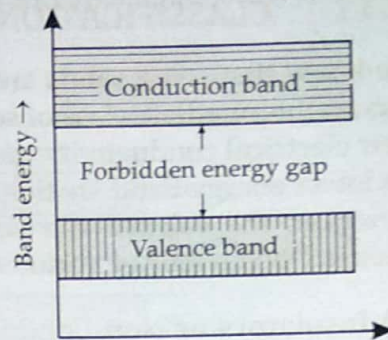


Fig. 9.19 Energy bands in solids.

9.11 CLASSIFICATION OF SOLIDS ON THE BASIS OF ENERGY BANDS

We know that some solids are good conductors of electricity while others are insulators. There is also an intermediate class of semiconductors. The difference in the behaviour of solids as regards their electrical conductivity can be beautifully explained in terms of energy bands. The electrons in the lower energy band are tightly bound to the nucleus and play no part in the conduction process. However, the valence and conduction bands are of particular importance in ascertaining the electrical behaviour of various solids.

(a) Insulators or Non-Conductors or Dielectrics

Insulators (e.g., wood, glass etc.) are those substances which do not allow the passage of electric current through them. In terms of energy band, the valence band is completely filled while the conduction band is empty. Further, the energy gap between valence and conduction bands is very large (≈ 15 eV) as shown in Fig. 9.20.

Therefore, a very high electric field is required to push the valence electrons to the conduction band. For this reason, the electrical conductivity of such materials is extremely small and may be regarded as nil under ordinary conditions.

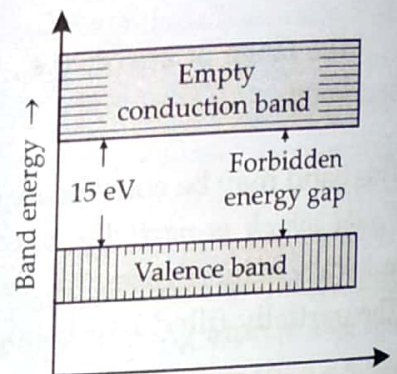


Fig. 9.20 Energy bands of insulators.

At room temperature, the valence electrons of the insulators do not have enough energy to cross over the conduction band. However, when the temperature is raised, some of the valence electrons may acquire enough energy to cross over to the conduction band. Hence the resistance of an insulator decreases with increase in temperature i.e., an insulator has negative temperature coefficient of resistance.

(b) Conductors

A second possibility is that of a solid in which the valence band is completely filled with electrons but there is no forbidden energy gap i.e., the valence and conduction bands overlap (Fig. 9.21). In such a solid, vacant energy levels are available adjacent to the filled levels. Therefore, it is possible to change the energy of the valence electrons by accelerating them by external electric field; the solid can carry a current. It will be a good conductor, for example: metals (copper, aluminium etc.)

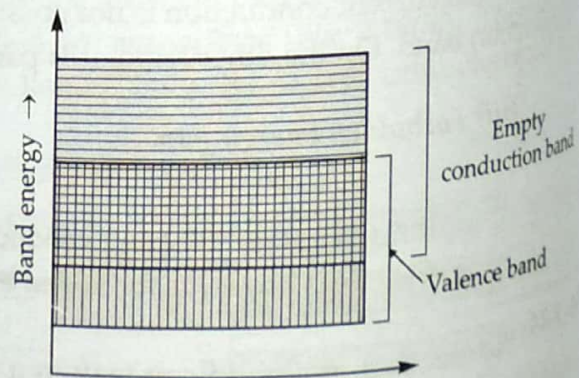


Fig. 9.21 Energy bands of metals.

The third possibility is that of a solid in which the valence band is only partially filled (Fig. 9.22), so that unfilled levels are available adjacent to the filled levels and external electronic

field is able to accelerate the valence electrons and make them carry a current. The solid is, thus a good conductor. This is a situation of alkali metals.

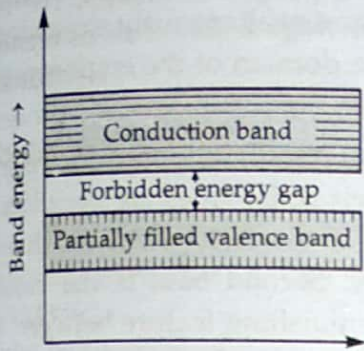


Fig. 9.22 Energy bands of alkali metals.

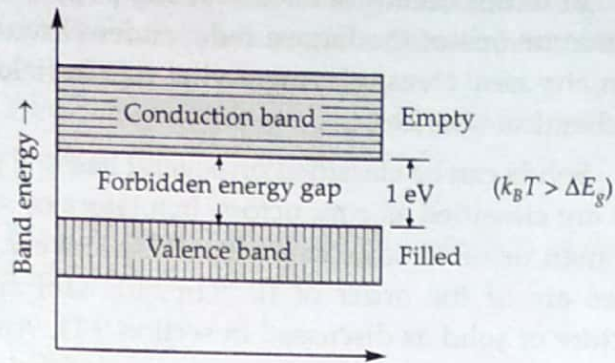


Fig. 9.23 Energy bands of intrinsic semiconductor

(c) Semiconductors

(i) **Intrinsic Semiconductors.** If the valence band is completely filled, conduction band is empty but the forbidden energy is small, of the order of about 1 eV (Fig. 9.23), the solid will be insulator at 0 K. At higher temperatures such that the thermal energy $k_B T$ is more than the gap energy ΔE_g , some of the valence electrons are able to jump over the gap into the conduction band. This creates free electrons in the conduction band and free holes in valence band. Thus the electron-hole pairs are thermally generated and can carry a small current through the solid. This is the case of an *intrinsic semiconductor*.

(ii) **Extrinsic Semiconductors.** In an *N-type extrinsic semiconductor*, the energy levels of the impurity atoms lie just below the bottom of the conduction band (Fig. 9.24). These electrons are very easily excited into the conduction band and are then capable of carrying a current through the solid.

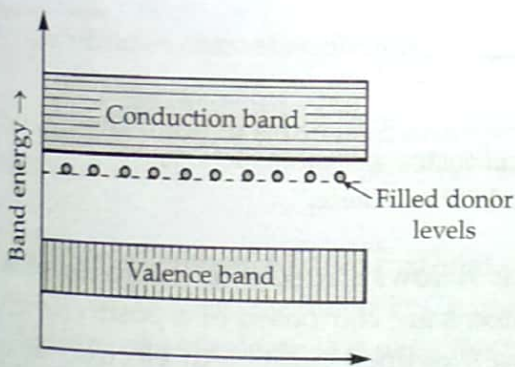


Fig. 9.24 N-type semiconductor.

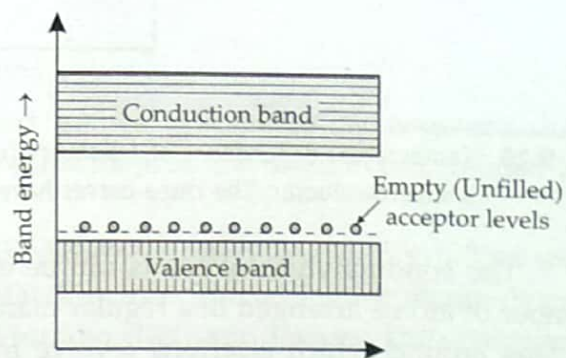


Fig. 9.25 P-type semiconductor.

Likewise, in a *P-type semiconductor*, the impurity atoms have unfilled energy levels, which lie just above the top of the valence band (Fig. 9.25). Electrons from the valence band are easily excited into these vacant levels leaving behind vacancies or free holes in the valence band which can carry a current through the solid.

9.12 SEMICONDUCTORS

In recent decade semiconducting physics has seen many dramatic new developments and has become one of the largest independent branches of physics. It has simultaneously expanded into many new areas, playing a vital role in fields that were once the domain of the engineering and chemical sciences.

Solids can be classified on several bases. One base is its electrical resistivity or conductivity. They are classified as conductors, insulators or semiconductors if their resistivities are very low, very high or of intermediate value respectively. The typical values of the resistivities for three classes are of the order of $10^{-8}\Omega\text{m}$, $10^{10}\Omega\text{m}$ and $1\Omega\text{m}$ respectively. Second base is the band structure of solid as discussed in section 9.11. Another essential distinguishing feature between a conductor and a semiconductor is the sign of their temperature coefficient of resistance. *Conductors* have a positive temperature coefficient of resistance at all temperatures and their resistance increases approximately linear with temperature.

An exceptional behaviour is shown by a *superconductor* for which the resistance sharply drops to very low values as the temperature is decreased below a critical temperature.

Semiconductors, on the other hand, have a negative temperature coefficient of resistance over an appreciable range of temperature the resistance decreasing with temperature exponentially.

This is shown in Fig. 9.26.

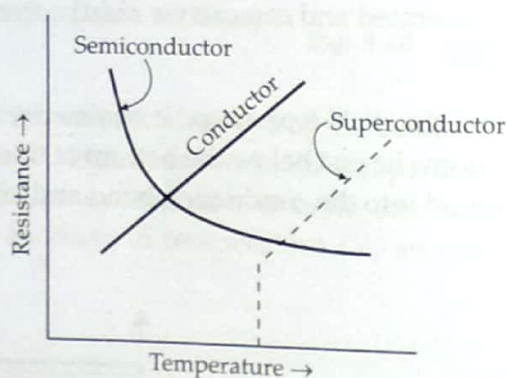


Fig. 9.26 Temperature dependence of resistance for a typical conductor, a semiconductor and a superconductor. The three curves have different scale of resistance.

The conductivities of solids can be explained as follows : A solid is made up of a large number of atoms arranged in a regular manner. The atoms are composed of a positively charged nucleus around which electrons revolve in certain well defined orbits. An electron in a orbit represents a bound electrons ; it is constrained to move in that particular orbit under the electrostatic attraction of the nucleus. Such an electron cannot carry current through the solid because it is not free to move along the direction of an external electric field. Therefore, a solid containing only the electrons that are bound in atoms will be an insulator. In order that a current might flow through the solid, it must contain some electrons which are free to move under the action of an external electric field without much hinderance. As we know that, a metal has a large number of such free electrons. Their movement in the absence of any electric field is random ;

as such it does not result in a non-zero average current in any direction. However, when electric field is applied, they acquire a non-zero average velocity in the direction of field and result in non-zero current through the solid. Thus a perfect insulator does not have any free electron, good conductor has a very large number of them and we expect that, semiconductor will have some of them.

9.13 CLASSIFICATION OF SEMICONDUCTORS

The classification of semiconductors on the basis of crystal structure and position of their constituent elements in the Periodic Table is shown in Fig. 9.27.

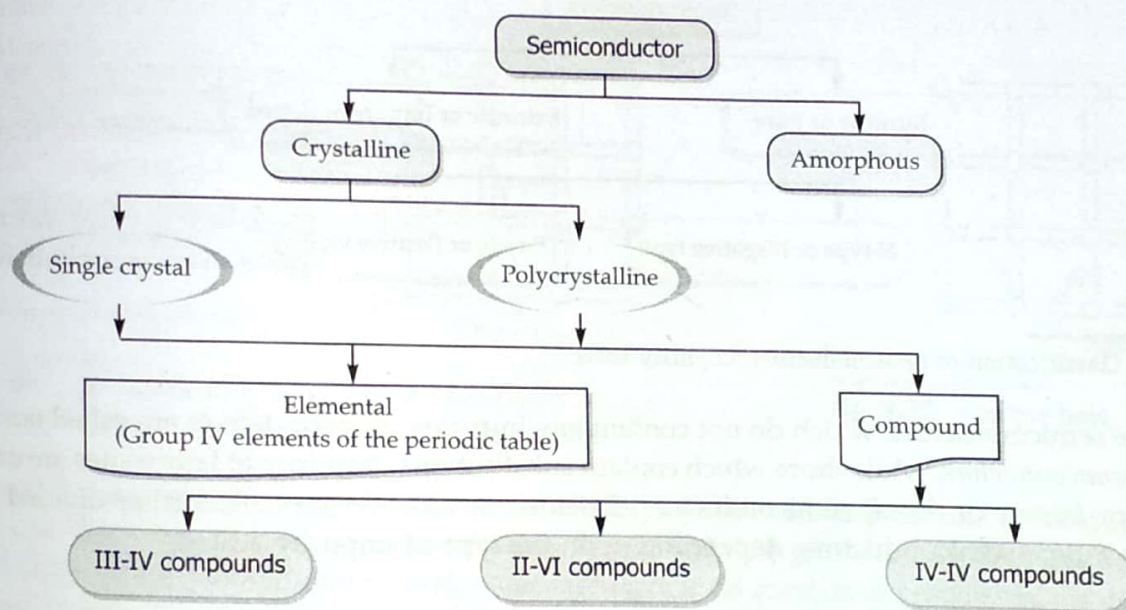


Fig. 9.27 Classification chart of semiconductors.

The classification of semiconductors produced much interest of scientists and engineers due to crystal structure and nature of bonding between the constituent atoms those in turn related to band gap.

Primarily, a semiconducting material is either crystalline or amorphous. The crystalline and amorphous materials are discussed in Chapter 7 – Crystal Structure. The crystalline materials are used in the form of a single crystal or in the polycrystal form in electronic devices. The crystalline semiconductors are further classified into single elemental material on the basis of their position in Periodic Table and compound semiconductors.

We know two most well known semiconductors of group IV i.e., silicon (Si) and germanium (Ge). These are said to be *elemental semiconductors*. Silicon and germanium both have four valence electrons and they crystallize in the diamond structure. If a semiconducting material has elements from two different groups or from the same group of the Periodic Table, it is known as a *compound semiconductor*. This compound semiconductor has important group i.e., III-IV compounds in which

one element from the third group and another element from the fifth group like gallium arsenide (GaAs), indium antimonide (InSb). On average they have four valence electrons. II-VI compounds are formed by combination of group II and group VI elements such as CdS and ZnO. The band gap of II-VI compounds is larger than III-V semiconductors. Another important group of semiconductors is IV-IV semiconducting compounds. This group of semiconductors is known as *narrow gap* semiconductors. Examples of IV-IV compounds are SiC, SiGe etc. These semiconductors are crystallized in simple rock salt type structure.

Semiconductors are also further divided in terms of purity of their crystal structures as shown in Fig. 9.28.

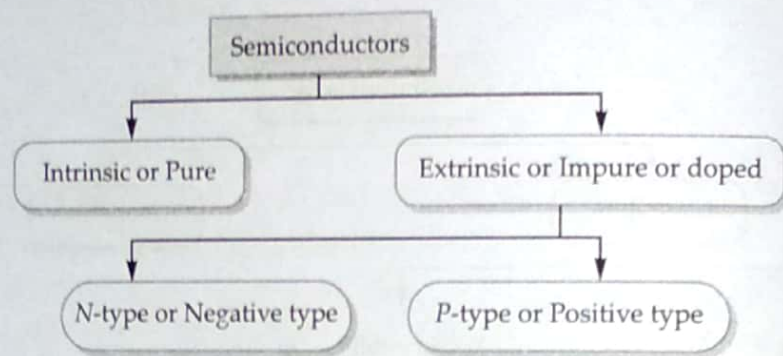


Fig. 9.28 Classification of semiconductors on purity basis.

The semiconductors, which do not contain any impurity atom or defects are called *intrinsic* (or *pure*) *semiconductors*, while those which contain substitutional impurity at lattice sites are called *extrinsic* (or *impure* or *doped*) *semiconductors*. Extrinsic semiconductors are further divided into *N-type* or *P-type* semiconductors, depending upon the type of impurity added.

9.14 INTRINSIC SEMICONDUCTORS

When the conductivity in crystalline semiconductor is only due to breaking of covalent bonds of the electrons in the valence band, then the substance is said to be an intrinsic semiconductor.

9.14.1 Electrons and Positive Holes in a Semiconductor

When a covalent bond of Si or Ge crystal breaks, then a free electron (n) moves from valence band to conduction band of a semiconductor and it leaves behind an unfilled electronic state in the valence band (Fig. 9.29). The absence of an electron in valence band is called positive hole (p). The word "hole" in a semiconductor therefore refers to the empty energy levels in valence band. So whenever there is an extra electron in the conduction band there is a simultaneous production of hole in the valence band [Fig. 9.29(a)]. The number of electrons in conduction band and simultaneously, number of holes in valence band will increase with increase of temperature. But for intrinsic semiconductor number of electrons in conduction band (n) and number of holes in valence band (p) are always same.

So mathematically, we can say

$$n = p$$

$\Rightarrow$

$$n = p = n_i$$

...(9.57)

where n_i is the number of charge carriers in intrinsic semiconductor. The positive hole, regarded as the active particle in the valence band, in the same way that the electron is considered to be active particle in conduction band. This leads to the concept of conductivity in semiconductors is caused not only by the motion of electrons in the conduction band which give rise to conduction current but also by the motion of the positive hole which gives rise to hole current in valence band. The movement of hole in the valence band is understood from Figs. 9.29–9.30. Hole positions can be changed from one place to other by interchanging their positions with electrons (Figs. 9.29, 9.30). These positive holes are also physical entities whose movement constitute a flow of current, which is called *hole current*.

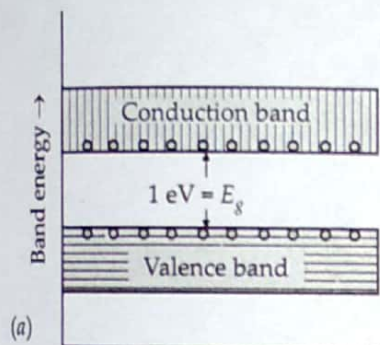


Fig. 9.29 Formation of holes in V.B. and movement of holes in V.B. which give rise to hole-current.

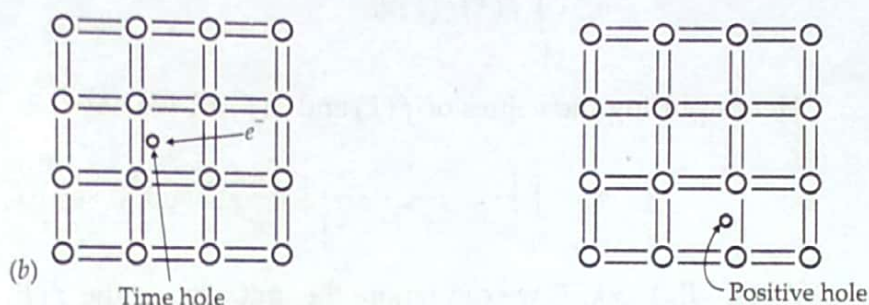


Fig. 9.30 Positive hole

Since movement of hole is opposite to that of electron, so the direction of hole current is same as direction of current flow. So in semiconductors, there are two charge carriers, electron and hole. But in metals electrons are the only charge carriers, because there is no question of breaking bonds and thereby forming the holes.

9.14.2 Carrier Concentration and Fermi Level

We have assumed here that the width of the allowed energy band is comparable to the forbidden gap of width E_g . Conduction band electrons have energy lying between E_C and ∞ , and, Valence band electrons have energy lying between $-\infty$ to E_V .

This is shown in Fig. 9.31.

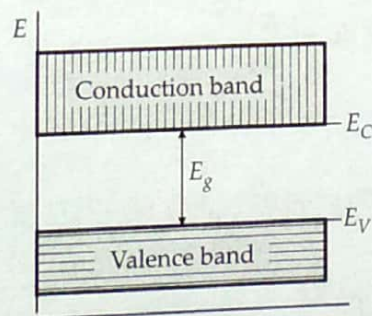


Fig. 9.31 Energy band diagram of an intrinsic semiconductor.

(a) Electron Concentration in Conduction Band

We know the total numbers of electrons per unit volume in conduction band is given by in terms of Fermi-Dirac distribution function $f(E)$ and density of state function $g(E)dE$ as

$$n_e = \int_{E_C}^{\text{top}} f(E)g(E)dE \quad \dots(9.58)$$

Here $f(E)$ decreases as we rise through the conduction band because $f(E)=0$ for $E \gg E_F$, So we have set the upper limit for energy as infinity for conduction band.

Hence, Eq. (9.58) becomes

$$n_e = \int_{E_C}^{\infty} f(E)g(E)dE$$

Hence putting the values of $f(E)$ and $g(E)dE$, we have

$$n_e = \int_{E_C}^{\infty} \left[\frac{1}{e^{(E-E_F)/k_B T} + 1} \right] \times \left[\frac{4\pi}{h^3} (2m_e^*)^{3/2} \times (E-E_C)^{1/2} dE \right]$$

As $(E-E_F) \gg k_B T$, we can ignore the unit term in the $f(E)$ and so

$$\begin{aligned} n_e &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{\infty} [(E-E_C)^{1/2} \times e^{-(E-E_F)/k_B T}] dE \\ \Rightarrow n_e &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{\infty} (E-E_C)^{1/2} \times e^{-(E-E_C)/k_B T} \times e^{-(E_C-E_F)/k_B T} dE \quad \dots(9.59) \end{aligned}$$

Let $y = (E-E_C)/k_B T$

$\Rightarrow dE = k_B T dy$ and $[E-E_C]^{1/2} = [y]^{1/2} [k_B T]^{1/2}$

Putting these into Eq. (9.59), we have

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} [k_B T]^{1/2} \times e^{-(E_F-E_C)/k_B T} \int_{E_C}^{\infty} (y)^{1/2} \times e^{-y} \times (k_B T) dy \quad \dots(9.60)$$

We know $\int_0^{\infty} y^{1/2} \times e^{-y} dy = \left(\frac{\pi}{4}\right)^{1/2}$

Hence, from Eq. (9.60)

$$\begin{aligned} n_e &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} [k_B T]^{1/2} \times e^{-(E_F-E_C)/k_B T} \times \left(\frac{\pi}{4}\right)^{1/2} \\ &= 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \times e^{-(E_F-E_C)/k_B T} \quad \dots(9.61) \end{aligned}$$

(a) Electron Concentration in Conduction Band

We know the total numbers of electrons per unit volume in conduction band is given by in terms of Fermi-Dirac distribution function $f(E)$ and density of state function $g(E)dE$ as

$$n_e = \int_{E_C}^{\text{top}} f(E)g(E)dE \quad \dots(9.58)$$

Here $f(E)$ decreases as we rise through the conduction band because $f(E)=0$ for $E \gg E_F$, So we have set the upper limit for energy as infinity for conduction band.

Hence, Eq. (9.58) becomes

$$n_e = \int_{E_C}^{\infty} f(E)g(E)dE$$

Hence putting the values of $f(E)$ and $g(E)dE$, we have

$$n_e = \int_{E_C}^{\infty} \left[\frac{1}{e^{(E-E_F)/k_B T} + 1} \right] \times \left[\frac{4\pi}{h^3} (2m_e^*)^{3/2} \times (E-E_C)^{1/2} dE \right]$$

As $(E-E_F) \gg k_B T$, we can ignore the unit term in the $f(E)$ and so

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{\infty} [(E-E_C)^{1/2} \times e^{(E_F-E)/k_B T}] dE$$

$$\Rightarrow n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} \int_{E_C}^{\infty} (E-E_C)^{1/2} \times e^{(E_F-E_C)/k_B T} \times e^{(E_C-E)/k_B T} dE \quad \dots(9.59)$$

$$\text{Let } y = (E-E_C)/k_B T$$

$$\Rightarrow dE = k_B T dy \text{ and } [E-E_C]^{1/2} = [y]^{1/2} [k_B T]^{1/2}$$

Putting these into Eq. (9.59), we have

$$n_e = \frac{4\pi}{h^3} (2m_e^*)^{3/2} [k_B T]^{1/2} \times e^{(E_F-E_C)/k_B T} \int_{E_C}^{\infty} (y)^{1/2} \times e^{-y} \times (k_B T) dy \quad \dots(9.60)$$

$$\text{We know } \int_0^{\infty} y^{1/2} \times e^{-y} dy = \left(\frac{\pi}{4} \right)^{1/2}$$

Hence, from Eq. (9.60)

$$\begin{aligned} n_e &= \frac{4\pi}{h^3} (2m_e^*)^{3/2} [k_B T]^{1/2} \times e^{(E_F-E_C)/k_B T} \times \left(\frac{\pi}{4} \right)^{1/2} \\ &= 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \times e^{(E_F-E_C)/k_B T} \quad \dots(9.61) \end{aligned}$$

or

$$n_e = N_C \exp\left[-\frac{E_C - E_F}{k_B T}\right] \quad \dots(9.62)$$

where $N_C = 2 \left[\frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2}$

Equation (9.61) or (9.62) gives the electron concentration in the conduction band.

(b) Hole Concentration in Valence Band (VB)

Here the probability for a state of energy E to be unoccupied is given by $[1 - f(E)]$.

Thus the total numbers of holes per unit volume is given by

$$n_h = \int_{\text{bottom}}^{E_V} [1 - f(E)] g(E) dE \quad \dots(9.63)$$

As earlier, we can change the lower limit as $-\infty$

Now

$$1 - f(E) = 1 - \frac{1}{e^{(E-E_F)/k_B T} + 1} = \frac{e^{(E-E_F)/k_B T}}{e^{(E-E_F)/k_B T} + 1}$$

However, for valence band, $E < E_F$, we have

$$e^{(E-E_F)/k_B T} \ll 1$$

Hence

$$1 - f(E) \cong e^{(E-E_F)/k_B T} \quad \dots(9.64)$$

Thus, in valence band, $[1 - f(E)]$ decreases exponentially as we move downwards, from top of the valence band. Now using Eq. (9.64) into Eq. (9.63), we have

$$n_h = \int_{-\infty}^{E_V} e^{(E-E_F)/k_B T} \times \left[\frac{4\pi}{h^3} (2m_e^*)^{3/2} (E_V - E)^{3/2} dE \right]$$

On solving as in above Eq. (9.60), we have

$$n_h = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2} \times e^{(E_V - E_F)/k_B T} \quad \dots(9.65)$$

$$n_h = N_V \exp\left[\frac{E_V - E_F}{k_B T}\right] \quad \dots(9.66)$$

where $N_V = 2 \left[\frac{2\pi m_h^* k_B T}{h^2} \right]^{3/2}$

Equation (9.65) or (9.66) gives the hole concentration in the valence band.

(c) Law of Mass Action

The law of mass action states that the product of the electron concentration in the conduction band and hole concentration in the valence band is equal to the square of intrinsic carrier concentration of the material. Let n_i , n_e and n_h are the intrinsic concentration of the material, the electron concentration in the conduction band and the hole concentration in the valence band respectively, then

$$n_i^2 = n_e n_h$$

Equation (9.67) is known as the *law of mass action*.

(d) Intrinsic Carrier Concentration

From the law of mass action $n_i^2 = n_e \cdot n_h$

Substituting the values of n_e and n_h given by Eqs. (9.62) and (9.66)

$$n_i^2 = N_C N_V \exp\left[\frac{E_F - E_C}{k_B T}\right] \exp\left[\frac{E_V - E_F}{k_B T}\right]$$

or

$$n_i^2 = N_C N_V \exp\left[\frac{-(E_C - E_V)}{k_B T}\right]$$

Since $E_C - E_V = E_g$, the band gap of the semiconducting material, Eq. (9.68) may be written as

$$n_i^2 = N_C N_V \exp\left(\frac{-E_g}{k_B T}\right)$$

Substituting the values of N_C and N_V in Eq. (9.69), we get

$$\begin{aligned} n_i^2 &= 2 \left[\frac{2\pi m_e^* k_B T}{h^2} \right]^{3/2} \times 2 \left[\frac{2\pi m_h^* k_B T}{h^2} \right]^{3/2} \exp\left(\frac{-E_g}{k_B T}\right) \\ &= 4 \left[\frac{2\pi k_B T}{h^2} \right]^3 \times [m_e^* m_h^*]^{3/2} \exp\left(\frac{-E_g}{k_B T}\right) \end{aligned}$$

Multiplying the numerator and denominator of Eq. (9.70) by m^3 , we get

$$n_i^2 = 4 \left[\frac{2\pi k_B m T}{h^2} \right]^3 \times \left[\frac{m_e^* m_h^*}{m^2} \right]^{3/2} \exp\left(\frac{-E_g}{k_B T}\right)$$

The value of n_i can be written as

$$\begin{aligned} n_i &= 2 \left[\frac{2\pi k_B m T}{h^2} \right]^{3/2} \times \left[\frac{m_e^* m_h^*}{m^2} \right]^{3/4} \exp\left(\frac{-E_g}{2 k_B T}\right) \\ &= 2 \left[\frac{2\pi k_B m}{h^2} \right]^{3/2} \times \left[\frac{m_e^* m_h^*}{m^2} \right]^{3/4} T^{3/2} \exp\left(\frac{-E_g}{2 k_B T}\right) \end{aligned} \quad \dots(9.72)$$

Equation (9.72) may also be written as

$$n_i = A T^{3/2} \exp\left[\frac{-E_g}{2 k_B T}\right] \quad \dots(9.73)$$

where $A = 2 \left[\frac{2\pi k_B m}{h^2} \right]^{3/2} \times \left[\frac{m_e^* m_h^*}{m^2} \right]^{3/4} = 4.82 \times 10^{21} \times \left[\frac{m_e^* m_h^*}{m^2} \right]^{3/4}$

Equation (9.72) gives the intrinsic carrier concentration in semiconductor.

(e) Value of Fermi Energy

In intrinsic semiconductor $n_e = n_h$

So equating Eqs. (9.61) and (9.65), we have

$$\begin{aligned} 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \times e^{(E_F - E_C)/k_B T} &= 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2} \times e^{(E_V - E_F)/k_B T} \\ \Rightarrow (m_e^*)^{3/2} \times e^{(E_F - E_C)/k_B T} &= (m_h^*)^{3/2} \times e^{(E_V - E_F)/k_B T} \\ \Rightarrow e^{(2E_F - E_C - E_V)/k_B T} &= \left(\frac{m_h^*}{m_e^*} \right)^{3/2} \end{aligned} \quad \dots(9.74)$$

$$\begin{aligned} \Rightarrow \left\{ \frac{2E_F - E_C - E_V}{k_B T} \right\} &= \frac{3}{2} \log \left(\frac{m_h^*}{m_e^*} \right) \\ \Rightarrow E_F &= \frac{E_C + E_V}{2} + \frac{3}{4} k_B T \log \left(\frac{m_h^*}{m_e^*} \right) \end{aligned} \quad \dots(9.75)$$

$$\left[\text{for } m_e^* = m_h^*, \log \frac{m_h^*}{m_e^*} = \log 1 = 0 \right]$$

$$\Rightarrow E_F = \frac{E_C + E_V}{2} \quad \dots(9.76)$$

So Fermi level lies exactly in the *middle* of the top of valence band and the bottom of the conduction band.

NOTE

Show that for an intrinsic semiconductor, Fermi level lies at the middle of the band gap.

Now putting the values of E_F into Eqs. (9.61) and (9.65) separately, we have

$$\begin{aligned} n_e &= 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \times e^{(E_F - E_C)/k_B T} \\ &= 2 \left(\frac{2 \pi k_B T}{h^2} \right)^{3/2} \times [(m_e^*)^2]^{3/4} e^{(E_C + E_F)/2 - E_C/k_B T} \end{aligned}$$

From Eq. (9.74),

$$\begin{aligned} n_e &= 2 \left(\frac{2 \pi k_B T}{h^2} \right)^{3/2} [(m_e^*)(m_h^*)]^{3/4} e^{(E_V - E_C)/2k_B T} \\ &= 2 \left(\frac{2 \pi k_B T}{h^2} \right)^{3/2} [m_e^* m_h^*]^{3/4} e^{-E_g/2k_B T} \end{aligned} \quad \dots(9.7)$$

and

$$n_h = 2 \left(\frac{2 \pi k_B T}{h^2} \right)^{3/2} [m_e^* m_h^*]^{3/4} e^{-E_g/2k_B T} \quad \dots(9.8)$$

Hence $n_h = n_e = n_i$, the intrinsic carrier concentration of the semiconductor.

Example 9.4 In a certain semiconductor, the mobility of electrons is $0.3 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$. The mobility of holes is $0.2 \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$ and forbidden energy band is 0.7 eV . Calculate the intrinsic carrier concentration of the semiconductor at 300 K , if effective mass of electrons and holes are respectively 0.55 and 0.37 times the mass of the electron.

Solution. Given : $T = 300 \text{ K}$, $E_g = 0.7 \times 1.6 \times 10^{-19} \text{ J}$, $m_e^* = 0.55 m_e = 0.55 \times 9.1 \times 10^{-31} \text{ kg}$, $m_h^* = 0.37 m_e = 0.37 \times 9.1 \times 10^{-31} \text{ kg}$, $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$ and $h = 6.626 \times 10^{-34} \text{ Js}$

$$\begin{aligned} \text{The intrinsic concentration } n_i &= 2 \left\{ \frac{2 \pi k_B T}{h^2} \right\}^{3/2} (m_e^* m_h^*)^{3/4} \exp \left[\frac{-E_g}{2 k_B T} \right] \\ &= 2 \left\{ \frac{2 \times \pi \times 1.38 \times 10^{-23}}{(6.626 \times 10^{-34})^2} \right\}^{3/2} [(0.55 \times 9.1 \times 10^{-31}) \times (0.37 \times 9.1 \times 10^{-31})] \exp \left[-\frac{0.7 \times 1.6 \times 10^{-19}}{2 \times 1.38 \times 10^{-23} \times 300} \right] \\ &= 1.35 \times 10^{13} \text{ m}^{-3} \end{aligned}$$

9.14.3 Extrinsic Semiconductors

The electrical conductivity of intrinsic semiconductor is very small. To increase the conductivity of intrinsic semiconductor a small percentage of trivalent or pentavalent atoms (impurity) are added to the pure semiconductor in the process of crystallification, which is called *doping* and results the impure semiconductor being called *extrinsic semiconductor*. The conductivity of extrinsic semiconductor is much higher, say for example 12 times than intrinsic semiconductor when an impurity is added 1 part in 10^8 . The impurity atom has a size which is almost of the same order of the host lattice. Since percentage of impurity atoms is very small, every impurity atom is surrounded by a normal lattice site. So basic structure of crystal will not get altered after doping.

There are two types of extrinsic semiconductors :

- N-type semiconductor
- P-type semiconductor.

(a) Negative Type or N-type Semiconductor

If a pentavalent atom is introduced in the intrinsic semiconductor say germanium the situation will as shown in Fig. 9.32 The impurity pentavalent atoms, some of the germanium atoms in the original crystal lattice and four of five valence electrons of the impurity atom will occupy four covalent bonds and fifth one will be almost free, as it does not find a place in the covalent bond. Every impurity atom thus contributes almost free electrons without creating a hole, and those free electrons act as carrier of current. A very small amount of energy is required to free the fifth electron from the atom, of the order of 0.01 eV for Ge and 0.05 eV for Si. Since those pentavalent impurities donate excess electron carrier, so they are called N-type impurity, and the crystal doped with donor impurity is called N-type semiconductor. Each ionized donor atom has a net charge of (+e) and is bound or immobile (Fig. 9.32).

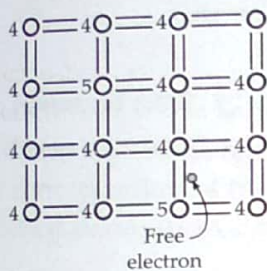


Fig. 9.32

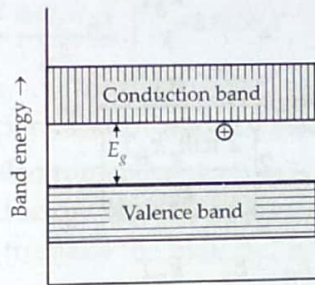


Fig. 9.33

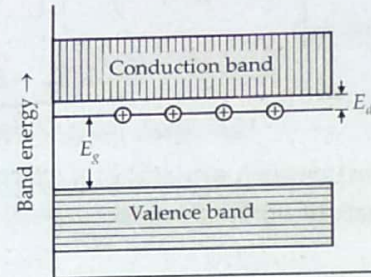


Fig. 9.34

When a donor impurities are doped to an intrinsic semiconductor Ge or Si, then additional discrete levels are introduced just below the conduction band in the forbidden gap (Fig. 9.34). These new additional level will be created and is known as *donor level* E_d (Fig. 9.34) because of additional impurity atoms are situated far apart in the crystal structure and hence their interaction is small. Since very small amount of energy (0.01 eV for Ge and 0.05 eV for Si) is required to free the electron from donor level to conduction band, therefore almost all the fifth electron of donor materials are raised to the conduction band at room temperature.

Fermi level corresponds to the centre of gravity of the electrons and holes. In the case of intrinsic semiconductor, the number of electrons are equal to number of holes. The *Fermi level* lies midway between the valence band and conduction band. In N-type semiconductor the Fermi level is lifted towards the conduction band as the conduction electrons are the majority carriers.

For N-type semiconductor a donor level E_d is formed below the Fermi level. At a temperature T , the density of conduction electron from Eq. (9.61) is given by

$$n_e = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right) e^{(E_F - E_C)/k_B T} \quad \dots(9.79)$$

The density of vacancies in the donor levels of energy E_d is given by

$$N_d^+ = n_d [1 - f(E)] = n_d \left[1 - \frac{1}{e^{(E_d - E_F)/k_B T} + 1} \right] = \left[\frac{e^{(E_d - E_F)/k_B T}}{e^{(E_d - E_F)/k_B T} + 1} \right]$$

Here N_d^+ is concentration for ionised donor atoms S . Assuming the position a little above the donor level by a few $k_B T$ the above expression can be approximated

$$N_d^+ = n_d e^{E_d - E_F / k_B T} \quad \dots(9.80)$$

However, the electron concentration in the conduction band and the vacancy concentration in donor level should be equal. Hence equating Eqs. (9.61) and (9.80), we have

$$n_e = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right) e^{(E_F - E_C)/k_B T} = n_d e^{E_d - E_F / k_B T}$$

$$\text{or } \log \left[2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \right] \left\{ \frac{E_F - E_C}{k_B T} \right\} = \log n_d + \frac{E_d - E_F}{k_B T}$$

$$\Rightarrow \frac{2 E_F - E_C - E_d}{k_B T} = \log \frac{n_d}{2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2}}$$

$$\Rightarrow E_F = \frac{E_C - E_d}{2} + \frac{k_B T}{2} \log \frac{n_d}{2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2}} \quad \dots(9.81)$$

At absolute zero temperature, $E_F = \frac{E_C + E_d}{2} \Rightarrow$ the Fermi level lies at half way between donor level and the bottom of the conduction band.

With an increase in T , E_F decrease and at room temperature it comes below the donor level.

Substituting the value of E_F from Eq. (9.81) into Eq. (9.61) we have the free electron concentration in the conduction band as

$$n_e = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \exp \frac{E_F - E_C}{k_B T}$$

$$= 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \exp \left[\frac{\frac{E_C + E_d}{2} + \frac{k_B T}{2} \log \frac{n_d}{2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2}} - E_C}{k_B T} \right]$$

$$\begin{aligned}
&= 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \exp \frac{E_d - E_C}{k_B T} + \frac{1}{2} \log \frac{n_d}{\left[2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \right]} \\
&= 2 \times \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} e^{(E_d - E_C)/2k_B T} \times \frac{n_d}{\left[2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{2/3} \right]} \\
&= (2 n_d)^{1/2} \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/4} e^{(E_d - E_C)/2k_B T}
\end{aligned}$$

The ionization energy is given by $\Delta E = E_C - E_d$ of the donors, then, we have

$$n_e = (2 n_d)^{1/2} \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/4} e^{-\Delta E/2k_B T} \quad \dots(9.82)$$

Equation (9.82) gives the carrier concentration of electrons in an N-type materials.

From Eq. (9.82), we say that electron concentration is proportional to the square root of the donor concentration of the semiconductor. With more increase E_F reaches up to the middle of the conduction band and valence band to make the material intrinsic.

Example 9.5 In an N-type semiconductor, the Fermi level lies 0.3 eV below the conduction band at 300 K. If the temperature is increased to 330 K, find the new position of the Fermi level.

Solution. Given $(E_C - E_F) = 0.3$ eV or $(E_F - E_C) = -0.3$ eV at temperature 300 K

$(E_C - E_F) = ?$ at temperature 330 K

We know that
$$n_e = 2 \left(\frac{2 \pi m_e^* k_B T}{h^2} \right)^{3/2} \exp \left(\frac{E_F - E_C}{k_B T} \right)$$

Assuming that the density of electrons in the conduction band of intrinsic semiconductor (n_e) remains unchanged by changing temperature 300 K to 330 K

$$\begin{aligned}
(n_e)^{300K} &= (n_e)^{330K} \\
\exp \left[\frac{E_F - E_C}{k_B \times 300} \right] &= \exp \left[\frac{E_F - E_C}{k_B \times 330} \right] \\
\Rightarrow \frac{0.3}{300 \text{ K}} &= \frac{(E_C - E_F)_{330}}{330 \text{ K}}
\end{aligned}$$

$$(E_C - E_F) \text{ at } 330K = \frac{0.3 \times 330}{300} = 0.33 \text{ eV}$$

Thus at 330 K, the Fermi energy level lies 0.33 eV below the conduction band.

(b) Positive Type or P-Type Semiconductor

When a trivalent impurity (boron $Z=5$, gallium $Z=31$ or indium $Z=49$) is added to the intrinsic semiconductor, only three of the covalent bond can be filled up and the vacancy that exists in the fourth bond constitute a *hole* (Fig. 9.35) without making an electron free. These holes can accept electrons and thus make available positive carriers in the valence band. So these trivalent impurities are called *acceptor* because every trivalent atom creates a *hole*, which accept electron and the intrinsic semiconductor with acceptor impurity is called *P-type semiconductor*. Each acceptor atom has a net charge $(-e)$ and is bound or immobile (Fig. 9.36).

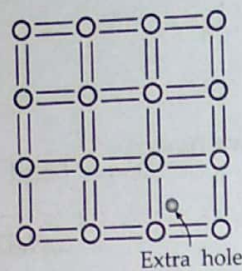


Fig. 9.35

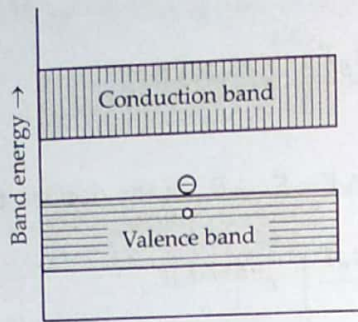


Fig. 9.36

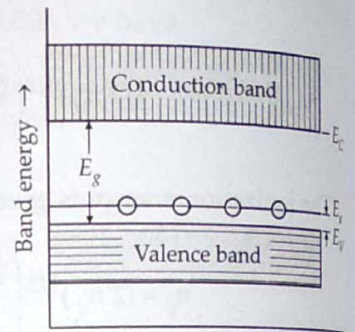


Fig. 9.37

When acceptor impurity is doped to an intrinsic semiconductor, Ge or Si, then the additional discrete energy levels are introduced. Just above the valence band (Fig. 9.37) in the forbidden gap which is called *acceptor level* (E_a). Since a very small amount of energy is necessary for an electron to leave the valence band and occupy the acceptor level, it thus creates a large number of holes in the valence band in the extrinsic semiconductor.

Thus by doping the impurity atoms in the intrinsic semiconductor not only conductivity is increased but it also serves to produce a conductor in which the electric carriers are mainly electrons or mainly holes.

For *P-type* semiconductors an acceptor level E_a is formed above the Fermi level and the valence band. At a temperature T , the density of holes is similar to Eq. (9.64) and is given by

$$n_h = 2 \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/2} e^{(E_V - E_F)/k_B T} \quad \dots (9.65)$$

NOTE

Electrons are majority carriers and holes are minority carriers in *N-type* semiconductor whereas in *P-type* semiconductor the majority carriers are holes and minority carriers are electrons.

The density of electrons in the acceptor levels of energy E_a is given by

$$N_a^- = n_a [f(E)] = n_a \frac{1}{e^{(E_a - E_F)/k_B T} + 1} \quad \dots (9.66)$$

Here N_a^- is ionized acceptor atoms concentration. However, the hole concentration in the valence band and the electron concentration in the acceptor level should be equal.

Also, assuming $E_a - E_F \gg a$ few $k_B T$, equating Eqs. (9.83) and (9.84), we have

$$2 \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/2} e^{(E_V - E_F)/k_B T} = n_a \frac{1}{e^{(E_a - E_F)/k_B T}}$$

On simplification

$$E_F = \frac{E_a + E_V}{2} - \frac{k_B T}{2} \log \frac{n_a}{\left[2 \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/2} \right]} \quad \dots(9.85)$$

At absolute zero temperature

$$E_F = \frac{E_a + E_V}{2}$$

$\Rightarrow$ The Fermi level lies at half way between acceptor level and the top of the valence band.

With an increase in T , E_F increases and at room temperature it comes above the acceptor level slightly. Hence we have

$$\Rightarrow n_h = (2 n_a)^{1/2} \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/4} e^{-\Delta E/k_B T} \quad \dots(9.86)$$

where $\Delta E = E_a - E_V$.

Equation (9.86) gives the carrier concentration of holes in a P-type material.

Example 9.6 In a P-type semiconductor, the Fermi level lies 0.4 eV above the valence band. If the concentration of the acceptor atom is tripled, find new position of the Fermi level.

Solution. Given P-type semiconductor, $E_F - E_V = 0.4$ eV, $n'_a = 3 n_a$,

also $k_B T = 0.03$ eV, $(E'_F - E_V) = ?$

For a P-type semiconductor, the hole density is given by

$$n_h = n_a = 2 \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/2} e^{(E_V - E_F)/k_B T} \quad \dots(i)$$

Similarly,

$$n'_h = n'_a = 3 n_a = 2 \left(\frac{2 \pi m_h^* k_B T}{h^2} \right)^{3/2} e^{(E_V - E'_F)/k_B T} \quad \dots(ii)$$

Comparing Eqs. (i) and (ii), we get

$$\begin{aligned} (E'_F - E_V) &= (E_F - E_V) - k_B T \ln 3 \\ &= (0.4 - 0.03 \times 1.098) \text{ eV} \\ &= (0.4 - 0.03294) \text{ eV} \\ &= 0.367 \text{ eV} \end{aligned}$$

9.15 FERMİ LEVEL : EFFECT OF TEMPERATURE ON INTRINSIC SEMICONDUCTOR

As we know that at absolute zero all the electronic states of the valence band are full and those of the conduction band are empty, so a semiconductor becomes an insulator at 0 K. But as the temperature increases some electrons from the valence band get sufficient energy and become free. They move to the conduction band and take part in conduction and give rise to conductivity in the semiconductor (Fig. 9.29).

Now we want to discuss the phenomenon with quantum mechanical point of view. Classically all electrons have zero energy at 0 K, but quantum mechanically electrons can not have zero energy at 0 K. The maximum energy that electrons may possess at 0 K is the Fermi energy (E_F). So quantum mechanically the electrons actually have energies extending from 0 to E_F at absolute zero temperature (Fig. 9.38).

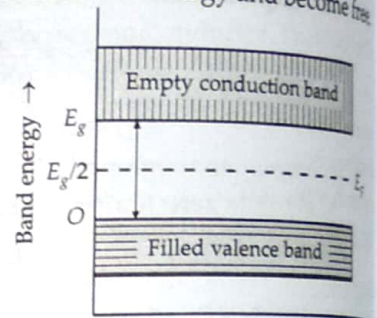


Fig. 9.38 Intrinsic semiconductor with Fermi level $E_F = (E_g/2)$.

Now in order to know how many of the electronic energy states in the valence band and conduction band will be occupied at different temperatures, we introduce a Fermi-factor $f(E)$ which is the number that expresses probability that a state of a given energy (E) is occupied by an electron under condition of thermal equilibrium. This number has a value between zero and unity and is a function of energy and temperature (or distribution function for Fermi-Dirac statistics) as

$$f(E) = \frac{1}{e^{(E-E_F)/(k_B T)} + 1} \quad \dots(9.87)$$

where k_B is Boltzmann constant, T is temperature in K and E_F is Fermi level of energy in eV, maximum energy that electron may possess at 0 K.

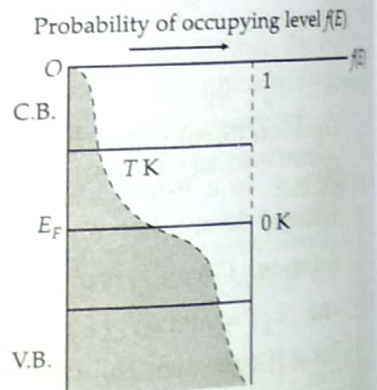


Fig. 9.39 Variation of Fermi factor $f(E)$ with temperature.

The Fermi factor is independent of the energy density of states, it is the probability that the states occupied at that level irrespective of the number of states actually present. That is, it is the fractional occupancy of possible states.

The variation of Fermi factor $f(E)$ with energy at different temperatures for intrinsic semiconductor is as shown in Fig. 9.39, where full line curve shows $f(E)$ at 0 K and dotted line curve at T K.

When $T = 0$ K; $f(E)$ can take two values

$$(i) \ E > E_F \quad f(E) = \frac{1}{1 + \exp(\infty)} = 0 \quad (\text{because } e^\infty = \infty)$$

$$(ii) \ E < E_F \quad f(E) = \frac{1}{1 + \exp(-\infty)} = 1 \quad (\text{because } e^{-\infty} = 0)$$

Thus at $T = 0\text{ K}$, $f(E) = 1$, when $E < E_F$, which means all the level below E_F i.e., valence band are filled up by electron and $f(E) = 0$, when $E > E_F$ i.e., all level above E_F i.e., conduction bands are empty. Because at 0 K , no heat energy is present, so no covalent bonds are being broken and the semiconductor behaves as an insulator.

But when $t = T\text{ K}$, then at $E = E_F$, $f(E)$ becomes

$$f(E) = \frac{1}{e^0 + 1} = \frac{1}{2} \quad [\because e^0 = 1] \quad \dots(9.90)$$

which means that when the temperature is not 0 K but some higher value say $T = 1000\text{ K}$, then some covalent bonds will break and some electrons will be available in conduction band whereas some electron vacancy i.e., hole will be available at valence band. This is shown by the dotted line in Fig. 9.39.

Example 9.7 Fermi energy of an intrinsic semiconductor is 0.6 eV . The low lying energy level in the conduction band is 0.2 eV above the Fermi level. Calculate the probability of occupation of this level by an electron at room temperature.

Solution. Given $E_F = 0.6\text{ eV}$, then $E = (0.6 + 0.2)\text{ eV} = 0.8\text{ eV}$

The probability of occupation of an energy level by an electron is given by the F-D distribution law

$$\begin{aligned} f(E) &= \frac{1}{1 + \exp \frac{E - E_F}{k_B T}} \\ &= \frac{1}{1 + \exp \frac{(0.8 - 0.6)\text{ eV} \times 1.6 \times 10^{-19}\text{ J}}{1.38 \times 10^{-23}\text{ JK}^{-1} \times 300\text{ K}}} = \frac{1}{1 + \exp 7.7} = 0.0004 = 0.04\% \end{aligned}$$

9.16 FERMI LEVEL IN EXTRINSIC SEMICONDUCTORS

In an intrinsic semiconductor the number of electrons is equal to number of holes ($n_i = p_i$). But in N-type extrinsic semiconductor number of electrons are increased due to doping of pentavalent atom ($n_e > n_i$) and number of holes are decreased ($n_h < n_i$) than which would be available for intrinsic semiconductor. The number of holes are decreased because of the large number of electrons present which increase the rate of recombination of electrons and holes. With similar argument it can be shown that in P-type extrinsic semiconductor the number of holes are increased due to doping of trivalent impurity atom ($n_h > n_i$) and number of electrons decreased ($n_e < n_i$) than which would be available for intrinsic semiconductor. Now since the Fermi level (E_F) is a measure of the probability of occupancy of the allowed energy states by the electron, so for an intrinsic semiconductor when $n_i = p_i$; Fermi level is at the centre of the forbidden gap [Fig. 9.40(a)]. But for N-type semiconductor, since $n_e > n_h$, it is clear that E_F must move closer to conduction band [Fig. 9.40(b)], to indicate that many of the energy states in the band are filled by the donor electrons, and of few holes exist in the valence band. Similarly, for P-type semiconductor, since $n_h > n_e$ so E_F must move from the centre of the forbidden gap closer to the valence band [Fig. 9.40(c)].

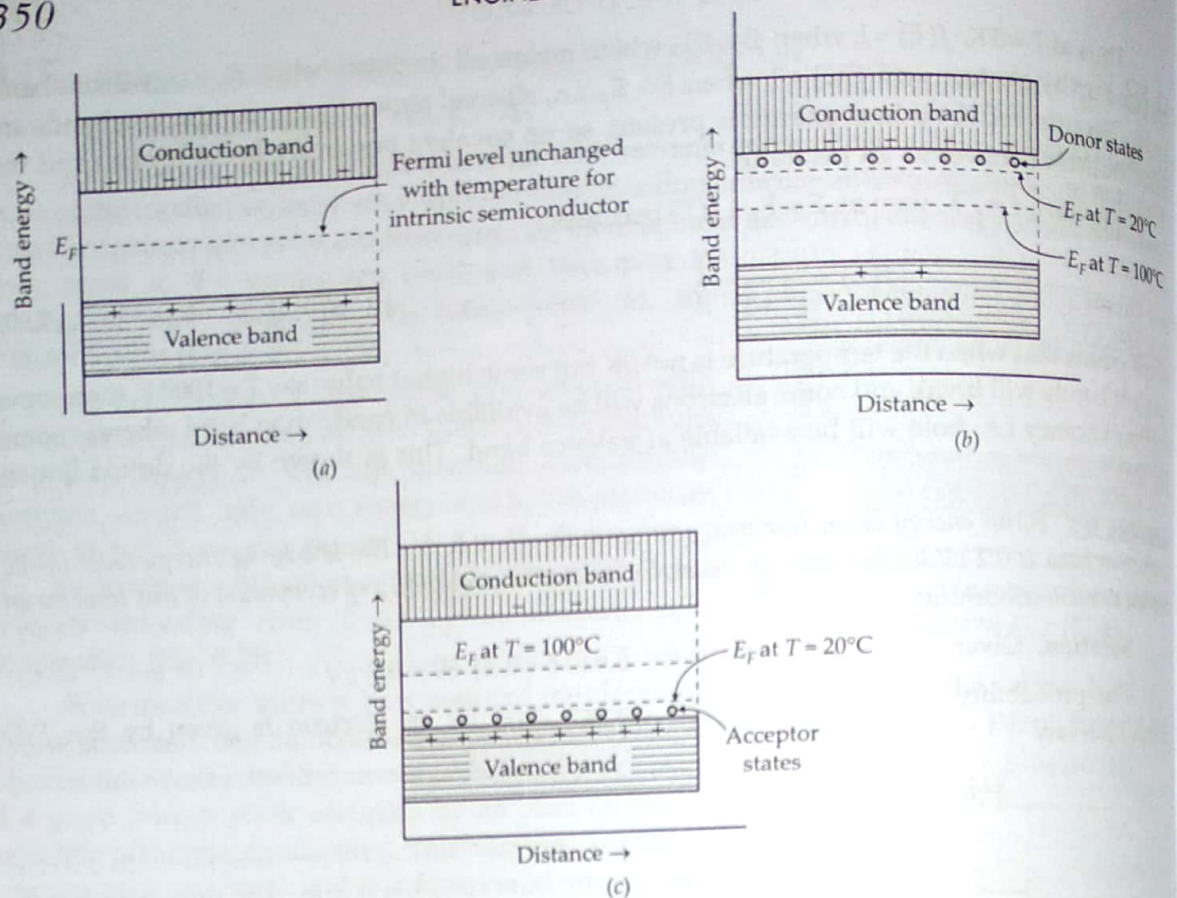


Fig. 9.40 Position of Fermi level (E_F) for (a) intrinsic semiconductor ;
(b) N -type semiconductor ; (c) P -type semiconductor.

9.17 VARIATION OF FERMI LEVEL WITH TEMPERATURE IN EXTRINSIC SEMICONDUCTORS

For an intrinsic semiconductor $n_i = p_i$ and as temperature increases both n_i and p_i will increase. Thus the Fermi level (E_F) will remain approximately at the centre of the forbidden gap. Thus for intrinsic semiconductor Fermi level is independent of temperature [Fig. 9.40(a)].

But in an extrinsic semiconductor it is different. Say for N -type material the electrons are coming from two different sources. Some electrons come from donor state, which are easily separated from their parent atom and they do not vary much as the temperature is increased. The other electrons in the conduction band are present because of the breaking of a covalent band. Such intrinsic effect will increase in number as temperature raised. So proportion of the total number of conduction electrons will increase from valence band. Thus as the temperature rises the material becomes more and more intrinsic and Fermi level moves closer to the intrinsic position, i.e., at the centre of the forbidden gap [Fig. 9.40(b)].

Similarly, for P -type semiconductor as temperature increases the material also becomes increasingly intrinsic and the Fermi-level rises until it approaches the centre of the gap as in the intrinsic semiconductor [Fig. 9.40(c)]. Thus both N -type and P -type materials become more and more like intrinsic material at high temperature. This places a limit on the operating temperature of an extrinsic semiconductor device.

9.18 THE HALL EFFECT

The Hall effect is the production of a potential difference (the Hall voltage) across an electrical conductor in which an electric current flows in the presence of a magnetic field. *Edwin Hall* discovered this effect in 1879. The conductivity measurements are not sufficient for the determination of the number of conducting charges n and their mobility μ . Moreover, these measurements do not give any information about the sign of the prominent charge carrier. The Hall effect supplies the information of the sign of charge carrier.

The phenomenon of Hall Effect is a consequence of motion of charge in both magnetic and electric fields. If a piece of conductor (metal or semiconductor) carrying current is placed in a transverse magnetic field, then an electric field or a potential difference is produced inside the conductor in a direction normal to both current and magnetic field. This phenomenon is known as *Hall effect*. The developed electric field and potential are called *Hall field* and *Hall voltage*.

Explanation

Consider a specimen in the form of a rectangular cross-section carrying current I_x in the x -direction. A uniform magnetic field B_z is applied along z -axis as shown in Fig. 9.41(a). If specimen is of conducting material or N -type semiconductor, then current flows due to motion of electron, but in P -type semiconductor and intrinsic semiconductor, current is carried by motion of holes and electrons-holes respectively.

A moving charge in transverse magnetic field experiences a force normal to both current and magnetic field. Thus moving charges slightly shift towards y direction due to magnetic

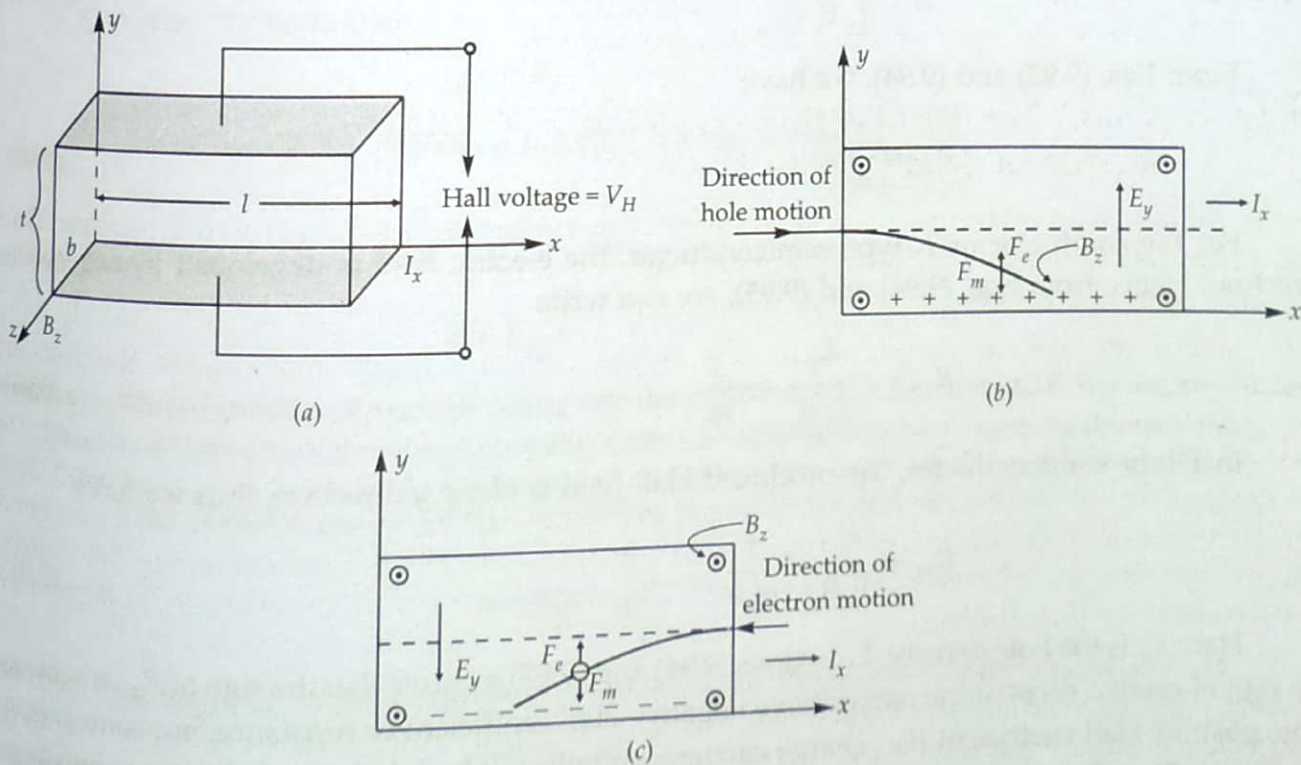


Fig. 9.41 (a) Illustration of Hall effect (b) motion of hole in P -type semiconductor specimen
(c) motion of electron of N -type semiconductor or metal specimen of Hall experiment

force $ev_x B_z$ [Figs. 9.41(a) and (c)]. This shift of charges produces a potential difference or electric field E_y along y -direction. The produced electric field also applies a force $e E_y$ on charges. When both the forces become equal, the equilibrium occurs, i.e., upto this condition charges are in equilibrium along y direction.

In equilibrium, $F_e = F_m$

$$e E_y = ev_x B_z$$

$$E_y = v_x B_z$$

If J_x is the current density in x direction, then

$$J_x = \frac{I_x}{A} = nev_x$$

Here n is density of charge carrier (electron or hole or both) and A (= breadth $\times$ thickness) is the cross-sectional area of specimen perpendicular to direction of J_x .

From Eqs. (9.91) and (9.92), we can write

$$E_y = \frac{B_z J_x}{ne}$$

This is the expression for Hall field. The Hall effect is described by means of Hall coefficient R_H , defined in terms of J_x by the following relation :

$$R_H = \frac{E_y}{J_x B_z}$$

From Eqs. (9.93) and (9.94), we have

$$R_H = \frac{1}{ne}$$

For the conductor or N -type semiconductor, the electric field is developed in negative y direction. Hence from Eqs. (9.94) and (9.95), we can write

$$R_H = -\frac{E_y}{J_x B_z} = -\frac{1}{ne}$$

In P -type semiconductor, the produced Hall field is along y direction, thus we have

$$R_H = \frac{E_y}{J_x B_z} = \frac{1}{n_h e}$$

Here n_h is the hole density. Equations (9.96) and (9.97) indicate that the sign of R_H is same as the sign of carrier. Most of the metals have negative Hall coefficient or resistance, but some metals have positive Hall coefficient (i.e., charge carriers are holes). If both holes and electrons contribute to current, then R_H can be positive or negative depending upon the relative density and mobility of carriers.

Experimentally the Hall coefficient is determined by the measurement of Hall voltage V_H which generates the Hall field. If $V_H = V_y$ is the Hall voltage across the thickness t of specimen, then

$$E_y = -\frac{V_y}{t} \quad \dots(9.98)$$

From Eqs. (9.92), (9.94) and (9.98), we have

$$R_H = \frac{V_y}{I_x} \times \frac{A}{B_z} \times \frac{1}{t} = \frac{V_y}{I_x} \times \frac{bt}{B_z} \times \frac{1}{t} \quad \dots(9.99)$$

$$R_H = \frac{V_y b}{I_x B_z}$$

The parameters at right hand side of Eq. (9.99) can be measured experimentally, and hence R_H can be calculated. The polarity of V_H will be opposite in *N*- and *P*-type semiconductors due to opposite electric field direction.

Mobility and Hall Angle

The mobility of charge carrier is defined as velocity per unit electric field, i.e.,

$$\mu = \frac{v_x}{E_x} \quad \dots(9.100)$$

Here E_x is the electric field due to which current flows along *x*-direction. If σ is conductivity of specimen material, then

$$J_x = \sigma E_x \quad \dots(9.101)$$

From Eqs. (9.99), (9.100) and (9.101), we can write

$$\mu = \frac{v_x \sigma}{J_x} = \frac{v_x \sigma}{ne v_x} = \frac{\sigma}{ne}$$

$$\mu = R_H \sigma \quad \dots(9.102)$$

Thus mobility of charge carrier can be determined if the values of R_H and σ are known. There is another importance parameter, with the help of which, μ can be determined.

That is called *Hall angle* θ_H . It is defined by the following relation :

$$\tan \theta_H = \frac{E_y}{E_x} \quad \dots(9.103)$$

Putting values from Eqs. (9.94), (9.101) and (9.102) in Eq. (9.103), we have

$$\tan \theta_H = \frac{E_y \sigma}{J_x} = \frac{E_y}{J_x} \mu$$

$$\tan \theta_H = \frac{E_y}{J_x} \times \frac{J_x B_z \mu}{E_y}$$

$$\tan \theta_H = \mu B_z$$

$$\mu = \frac{1}{B_z} \tan \theta_H$$

...(9.104)

Importance or Application of Hall Effect

1. The sign of Hall coefficient can be used to determine whether a given semiconductor is N or P-type.
2. The carrier concentration can be evaluated when R_H is known (as $n = 1/R_H e$).
3. Since $\mu = R_H \sigma$, thus mobility of charge carrier can also be well determined with the help of R_H .
4. Since Hall voltage is proportional to magnetic flux density B_z and current I_x through a sample or specimen. The Hall effect can be used as the basis for design of a magnetic flux density meter.
5. Hall effect can be used to determine the power flow in an electromagnetic wave because V_H is proportional to product of E_y and B_z , i.e., magnitude of Poynting vector in e.m. wave.

Example 9.8 A P-type semiconductor has Hall coefficient equal to $0.0125 \text{ m}^3/\text{C}$. Find out the density of charge carrier in it.

Solution. Given $R_H = 0.0125 \text{ m}^3/\text{C}$ and $e = 1.6 \times 10^{-19} \text{ C}$

$$R_H = \frac{1}{Ne}$$

$$N = \frac{1}{R_H e}$$

$$= \frac{1}{0.0125 \times 1.6 \times 10^{-19}} = 50 \times 10^{19} \text{ per m}^3$$

$$= 50 \times 10^{20} \text{ per m}^3$$

Example 9.9 A semi conducting specimen of size $10 \text{ mm} \times 5 \text{ mm} \times 2 \text{ mm}$ is used in Hall effect experiment. A current of 10 mA is flowing in it along its length. If 20 mV Hall voltage develops across its thickness, when it is placed in 0.5 T transverse magnetic field, then find Hall coefficient. [see Fig. 9.24(a)]

Solution. Given that $V_H = 20 \text{ mV} = 20 \times 10^{-3} \text{ V}$, $b = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$

$$I_x = 10 \text{ mA} = 10 \times 10^{-3} \text{ A} \text{ and } B_z = 0.5 \text{ T}$$

Since $R_H = \frac{V_H}{I_x} \times \frac{b}{B_z}$

$$R_H = \frac{20 \times 10^{-3} \times 5 \times 10^{-3}}{10 \times 10^{-3} \times 0.5} = 0.02 \text{ m}^3/\text{C}$$

Formulae at a Glance

9.1 Potential energy, $V = \frac{-Ze^2}{4\pi\epsilon_0 r}$

9.2 Bloch Theorem,
 $U_k(x) = U_x(x+a)$

9.3 The Kronig-Penney Model,
(i) $\cos k(a+b) = \frac{\alpha^2 - \beta^2}{2\alpha\beta} \sinh \alpha b \sin \beta a$
 $+ \cosh \alpha b \cos \beta a$

(ii) $\cos ka = \frac{E}{\beta a} \sin \beta a + \cos \beta a$

9.4 Brillouin zone

(i) 1-Dimensional, $k = \frac{n\pi}{a}$

(ii) 2-Dimensional,
 $n_1 k_x + n_2 k_y = (n_1^2 + n_2^2) \frac{\pi}{2}$

for $n_1 = \pm 1$ and $n_2 = 0$

$k_x = \pm \left(\frac{\pi}{2}\right)$

for $n_1 = 0$ and $n_2 = \pm 1$

$k_y = \pm \left(\frac{\pi}{2}\right)$

(iii) 3-Dimensional,

$k_x n_1 + k_y n_2 + k_z n_3 = (n_1^2 + n_2^2 + n_3^2) \frac{\pi}{a}$

For $n_1 = \pm 1$, $n_2 = n_3 = 0$

$k_x = \pm \frac{\pi}{a}$

For $n_1 = 0$, $n_2 = \pm 1$, $n_3 = 0$

$k_y = \pm \frac{\pi}{a}$

9.5 Effective mass of an Electron,

(i) $m^* = \frac{F_{ext}}{a}$

where F_{ext} = external force,
 a = acceleration.

(ii) $m^* = \hbar^2 \left[\frac{d^2 E}{dk^2} \right]^{-1}$

where E = electric field,
 k = wave vector

(iii) $E_F = \frac{\hbar^2}{2m^*} \left(\frac{3N}{8\pi V} \right)^{2/3}$

9.6 Law of mass action $n_i^2 = n_e n_h$

where

n_i = intrinsic concentration

n_e = electron concentration in
conduction band

n_h = hole concentration in valence band

9.7 Carrier concentration and Fermi level
(Intrinsic semiconductor)

(i) Electron concentration in conduction band

$n_e = 2 \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \exp (E_F - E_C) / k_B T$

(ii) Hole concentration in valence band

$n_h = 2 \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2} \exp (E_V - E_F) / k_B T$

(iii) Fermi energy

$E_F = \frac{E_C + E_V}{2}$

(iv) The intrinsic carrier concentration

$n_e = n_h = n_i = 2 \left(\frac{2\pi k_B T}{h^2} \right)^{3/2} (m_e^* m_h^*)^{3/4} \exp [-E_g / 2k_B T]$

9.8 Electron concentration in N-type semiconductor

$n_e = (2n_d)^{1/2} \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/4} \exp [-\Delta E / k_B T]$

[Here $\Delta E = E_C - E_d$]

9.9 Hole concentration in P-type semiconductor

$$p_h = (2n_i)^{1/2} \left(\frac{2\pi m_h^* k_B T}{h^2} \right)^{3/4} \exp \left[-\Delta E / k_B T \right]$$

[Here $\Delta E = E_g = E_V - E_C$]

9.10 Distribution Function for Fermi Dirac statistics

$$F(E) = \frac{1}{1 + \exp \left(\frac{E - E_F}{k_B T} \right)}$$

9.11 The Hall effect

(i) Hall coefficient

$$R_H = -\frac{1}{CNe} \text{ [in e.m.u.]}$$

$$R_H = -\frac{1}{Ne} \text{ [in e.s.u.]}$$

(ii) Mobility

$$\mu = R_H \sigma C \text{ (in e.m.u.)}$$

$$(iii) \tan \phi = \mu B_z \text{ or } \phi = \tan^{-1}(\mu B_z)$$

Miscellaneous Solved Numerical Problems

Problem 9.1 For a 2D square lattice of side $0.02 \text{ \AA}$, calculate the momentum of electron corresponding to the boundary of the first Brillouin zone.

Solution. We know the first Brillouin zone lies in the range $-\frac{\pi}{a} < k < \frac{\pi}{a}$ so momentum

electron

$$p = \hbar k = \hbar \left(\frac{\pi}{a} \right)$$

$$= \frac{1.05 \times 10^{-34} \text{ Js} \times 3.14}{0.02 \times 10^{-10} \text{ m}}$$

$$= 1.64 \times 10^{-22} \text{ kg ms}^{-1}$$

Problem 9.2 Show that for a square lattice (two dimensional) the kinetic energy of the free electron at corner of the first zone is higher than that of an electron at mid point of side face of zone by a factor 2.

Solution. Kinetic energy of the electron of momentum p is $p^2/2m$ and $p = \hbar k$

$$\text{vector } k = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} \text{ or } p = \hbar k$$

$$\text{and } K.E. = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

At the mid point of the side of the Brillouin zone $k = \pi/a$

$$E_1 = \frac{\hbar^2}{2m} \left(\frac{\pi}{a} \right)^2 = \frac{\hbar^2}{8am}$$

At the corner of the first zone,

$$k = \left[\frac{\pi^2}{a^2} + \frac{\pi^2}{a^2} \right]^{1/2} \quad \text{or} \quad k^2 = \frac{2\pi^2}{a^2}$$

$$\therefore \text{K.E. } E_2 = \frac{\hbar^2}{2m} \times \frac{2\pi^2}{a^2} = \frac{\hbar^2}{4am} = 2E_1$$

Problem 9.3 A 2D square lattice has side 2.5 Å. What will be the momentum of an electron whose wave terminates at the boundary of first Brillouin zone? Also calculate energy of the electron.

Solution. For first Brillouin zone $-\frac{\pi}{a} < k < \frac{\pi}{a}$, then momentum of electron

$$\begin{aligned} p &= \hbar k = \hbar \frac{\pi}{a} \\ &= \frac{1.05 \times 10^{-34} \text{ Js} \times 3.14}{2.5 \times 10^{-10} \text{ m}} = 1.38 \times 10^{-24} \text{ kg m/s} \end{aligned}$$

Corresponding value of the energy of electron

$$E = \frac{p^2}{2m} = \frac{(1.38 \times 10^{-24} \text{ kg m/s})^2}{2 \times (9.1 \times 10^{-31} \text{ kg})} = 1.04 \times 10^{-18} \text{ J} = 6.5 \text{ eV}$$

Problem 9.4 In an intrinsic semiconductor ($E_g = 0.676 \text{ eV}$), $m_e = 0.09 m$ and $m_h = 0.36 m$. Calculate the concentration of intrinsic charge carriers at 300 K.

Solution. $E_g = 0.676 \text{ eV}$, $m_e = 0.09 m$ and $m_h = 0.36 m$

Then carrier concentration

$$n_i = 2 \left(\frac{k_B T}{2\pi \hbar^2} \right)^{3/2} (m_e m_h)^{3/4} \exp \left(-\frac{E_g}{2k_B T} \right)$$

$$= CT^{3/2} (0.3 \times 0.6)^{3/2} \exp \left(-\frac{0.676}{0.052} \right)$$

$$C = 2 \left(\frac{k_B}{2\pi \hbar^2} \right)^{3/2} (9.1 \times 10^{-31})^{3/2} = 4.83 \times 10^{21} / \text{m}^3$$

$$T^{3/2} = (300)^{3/2} = 5.19 \times 10^3 \text{ K}$$

$$\therefore n_i = 4.83 \times 10^{21} \times (5.19 \times 10^3) \times (0.0763) \times (2.266 \times 10^{-6}) = 4.33 \times 10^{18} / \text{m}^3$$

Problem 9.5 For copper at 1000 K, find the energy at which the probability $f(E)$ that a conduction electron state will be occupied is 0.90.

Solution. The probability $f(E)$ of a state corresponding to energy E being occupied by an electron at temperature T is given by

$$f(E) = \frac{1}{1 + \exp \left(\frac{E - E_F}{k_B T} \right)} = \frac{1}{1 + e^{\Delta E / k_B T}}$$

Therefore
$$0.90 = \frac{1}{1 + e^{\Delta E / k_B T}}$$

or
$$\frac{\Delta E}{k_B T} = -2.20$$

Thus
$$\begin{aligned} \Delta E &= -2.20 k_B T \\ &= -2.20 \times (8.63 \times 10^{-5} \text{ eV / K}) \times (1000 \text{ K}) = -0.19 \text{ eV} \end{aligned}$$

For copper, $E_F = 7.00 \text{ eV}$ so that

$$E = E_F + \Delta E = 7.00 \text{ eV} - 0.19 \text{ eV} = 6.81 \text{ eV}$$

Problem 9.6 Hall voltage of 20 mV is found to be developed with a sample carrying a current of 15 mA is placed in a transverse magnetic field of 4 kilo gauss. Calculate the concentration of charge carriers in the sample. The thickness of the sample along the magnetic field direction is 0.4 mm.

Solution. Given Hall voltage $V_H = 20 \text{ mV}$, current $I = 15 \text{ mA}$,

$$B = 4 \text{ kilo gauss} = 0.4 \text{ Tesla, thickness } d = 0.4 \text{ mm} = 0.4 \times 10^{-3} \text{ m}$$

Hall coefficient
$$R_H = \frac{V_H d}{I_x B}$$

But
$$R_H = \frac{1}{ne} \text{ (numerically)}$$

Then concentration
$$\begin{aligned} n &= \frac{I_x B}{V_H d e} \\ &= \frac{15 \times 10^{-3} \times 0.4}{20 \times 10^{-3} \times 0.4 \times 10^{-3} \times 0.4 \times 10^{-19}} = 4.69 \times 10^{21} \text{ m}^{-3} \end{aligned}$$

Problem 9.7 InSb is an intrinsic semiconductor with energy gap of 0.15 eV. A Hall voltage of -4.07 mV is developed across 1 cm × 1 mm × 1 mm bar of InSb when a current $i_x = 0.1 \text{ A}$ passes along x-direction in the presence of magnetic induction 0.1 Wb/m² along z-direction. Calculate the Hall coefficient and density of carrier.

Solution.
$$E_H = V_H / 10^{-3} \text{ V.m}^{-1} = -4.07 \text{ Vm}^{-1}$$

and
$$\begin{aligned} R_H &= \frac{E_H}{J_x B_z} \\ &= \frac{4.07}{[(0.1/10^{-6}) \times 0.1]} \text{ m}^3 \text{C}^{-1} = 4.07 \times 10^{-4} \text{ m}^3 \text{C}^{-1} \end{aligned}$$

$$\therefore R_H = \frac{1}{ne}$$

$$n = \frac{1}{e R_H} = \frac{1}{4.07 \times 10^{-4} \times 1.6 \times 10^{-19}} \text{ m}^{-3} = 1.54 \times 10^{22} \text{ m}^{-3}$$

Problem 9.8 Calculate the Hall coefficient of sodium based on free electron model. Sodium has bcc structure and the side of the cube is $4.28 \text{ \AA}$.

Solution. Given that $a = 4.28 \text{ \AA} = 4.28 \times 10^{-10} \text{ m}$. Crystal is bcc

The number of electrons per unit volume for the sodium crystal is given by

$$n = \frac{2}{a^3} = \frac{2}{(4.28 \times 10^{-10})^3} = 2.55 \times 10^{28} / \text{m}^3$$

The Hall coefficient

$$R_H = \frac{1}{ne} = \frac{1}{2.55 \times 10^{28} \times 1.6 \times 10^{-19}} = 2.45 \times 10^{-9} \text{ m}^3 / \text{C}$$

Conceptual Questions

9.1 State the Bloch theorem.

Ans. The Bloch theorem is a mathematical statement regarding the form of the one electron wave functions for a perfectly periodic potential.

9.2 What is Kronig-Penney model ?

Ans. Kronig and Penney in 1931 solved the Schrödinger's equation for electrons in a simple idealised periodic field. The model is of considerable importance because it interprets the main features of the band structure of metals.

9.3 What is the basic assumption in Kronig-Penney model ?

Ans. In Kronig-Penney model, it is assumed that potential energy of an electron in a linear array of positive nuclei has the form a periodic array of square wells with period $(a + b)$.

At the bottom of the well, i.e., for $0 < x < a$ the electron is assumed to be in the vicinity of a nucleus and potential energy is taken as zero whereas outside a well i.e., for $-b < x < 0$ the potential energy is assumed to be V_0 as shown in Fig. 9.7.

9.4 What are Brillouin zones ?

Ans. If Schrödinger wave equation for electron energies is solved with a periodic function $u(k)$ to give the energies of an electrons in a solid, the solutions falls into permitted energy bonds. If the solutions are plotted in the reciprocal lattice of the crystal being considered, the zones enclosing the solutions for $k = 1, 2, 3, \dots, n$ are called Brillouin zones.

9.5 Define the effective mass of electron.

[GGSIPU, May 2014(2.5 marks)]

Ans. When an electron in a periodic potential is accelerated by an electric field or a magnetic field then the mass of the electron varies with velocity. This means the mass of the electron is a function of velocity and is termed as effective mass of the electron.

9.6 How is an energy band formed ?

Ans. When N number of atoms are brought together to form a solid, all the different discrete energy levels of electrons of the constituent atoms of the solid come close together to form a continuous band of energies. The Pauli's exclusion principle is not violated during the formation of an energy band.

9.7 Define energy band.

Ans. The range of energies possessed by the electrons in solid is known as energy band.

9.8 Where is Fermi energy level located in an intrinsic semiconductor ?

Ans. The Fermi level is located very close to the middle of the band gap slightly upwards towards the edge of conduction band due to difference of effective mass of electrons and holes.

9.9 How does a doped semiconductor behave with the increase of temperature ?

Ans. A doped extrinsic semiconductor tends to become an intrinsic one as temperature increases.

9.10 How does the Fermi level of doped semiconductor shifts its position with increase of temperature ?

Ans. As temperature increases the Fermi level of a doped semiconductor shifts towards the intrinsic level i.e., towards the midgap.

9.11 What is the hole ?

Ans. The vacant place due to absence of electron is called hole.

9.12 In doped semiconductor how does the free carrier concentration change with increase in temperature ?

Ans. In doped semiconductors there are three regions as far as the carrier concentration versus temperature variation is concerned.

- ❖ At low temperature the electron (hole) from the donor (acceptor) is bound to the dopant. This is the freezeout region.
- ❖ With increase of temperature the fraction of ionized donors or acceptors increases and ultimately all the donors and acceptors are ionized and the free carrier concentration becomes equal to the dopant concentration. This is called saturation region. Semiconductor devices are normally preferred to operate in the temperature range at which saturation occurs.
- ❖ As the temperature is further increased beyond saturation region, the carrier concentration again increases due to increases the intrinsic generation across the band gap and becomes larger than the doping concentration. This is called intrinsic region.

9.13 What do you mean by thermal equilibrium in a semiconductor ?

Ans. Equilibrium or thermal equilibrium in a semiconductor means that no external forces such as electric field, magnetic field, radiation or temperature gradient act on the semiconductor. All properties of the semiconductor will be independent of time in thermal equilibrium.

9.14 Where is Fermi energy level located in an intrinsic semiconductor ?

Ans. The Fermi level is located very close to the middle of the band gap slightly upwards towards the edge of the conduction band due to the difference of the effective mass of the electrons and holes.

9.15 Why do metals have a positive temperature coefficient of resistance and semiconductor have negative temperature coefficient of resistance ?

Ans. A large number of free electrons participate in conduction in metals. With increase of temperature these electrons are scattered more frequently by increasing number of phonons and the relaxation time for collision decreases. As a result mobility decreases and resistivity increases with the increase mobility decreases and resistivity increases with the increase of temperature in metals. In case of semiconductors, as temperature increases large number of electron-hole pairs are generated thermally. Therefore, conductivity increases and resistivity decreases.

9.16 How does a doped semiconductor behave with the increase of temperature ?

Ans. A doped or extrinsic semiconductor tends to become an intrinsic one as temperature increases.

9.17 How does the Fermi level of doped semiconductor shift its position with increase of temperature?

Ans. As temperature increases the Fermi level of a doped semiconductor shifts towards the intrinsic level i.e., towards the midgap.

9.18 What is the effect of dopant concentration on the Fermi energy and carrier concentration of N- and P-type semiconductors?

Ans. In N-type semiconductor, increase in dopant concentration increases the majority charge carrier concentration and the Fermi level move closer to the bottom of the conduction band.

In P-type semiconductor, increase in dopant concentration increases the majority charge carrier concentration and pushes the Fermi level closer to the top of the valence band.

9.19 Distinguish between intrinsic and extrinsic semiconductors.

[GGSIPU, May 2010 (2.5 marks)]

Ans.

Intrinsic Semiconductor	Extrinsic Semiconductor
1. It is a semiconductor in its pure form.	It is a semiconductor doped with impurities.
2. These semiconductors have low electrical conductivity.	These semiconductors have high electrical conductivity.
3. The operating temperature is low for this type of semiconductors.	The operating temperature is high for this type of semiconductors.
4. Charge carriers are produced due to thermal excitation. Examples : Ge, Si	Additional charge carriers are also produced due to added impurities. Examples : Ge and Si doped with P, As, Bi, Sb etc.

9.20 What is difference between N-type and P-type semiconductors?

[GGSIPU, April 2010-Reappear (2 marks)]

Ans.

N-type Semiconductor	P-type Semiconductor
1. These are materials, which have pentavalent impurity atoms (donors) added and conduct by "electron" movement and are called N-type semiconductors.	These are materials, which have trivalent impurity atoms (acceptors) added and conduct by "hole" movement are called P-type semiconductors.
2. In these types of materials : ■ The donors are positively charged. ■ These are large number of free electrons. ■ A small number of holes in relation to the number of free electrons. ■ Doping gives : - positively charged donors - negatively charged free electrons ■ Supply of energy gives : - negatively charged free electrons - positively charged holes.	In these types of materials : ■ The acceptors are negatively charged. ■ These are large number of holes. ■ A small number of free electrons in relation to the number of holes. ■ Doping gives : - negatively charged acceptors - positively charged holes. ■ Supply of energy gives : - positively charged holes - negatively charged free electrons.

Both P-and N-types as whole are electrically neutral.

9.21 What do you mean by Hall effect and Hall voltage ?

Ans. If a current carrying conductor is placed in a magnetic field, a potential difference is developed across the conductor in a direction perpendicular to both current and magnetic field. This phenomenon is called the Hall effect and the induced potential difference is called the Hall voltage.

9.22 What is the importance of Hall effect ?

Ans. The importance of Hall-effect is given below :

- (i) Unknown magnetic field can be estimated by the knowledge of Hall voltage.
- (ii) The nature of the current carriers can be identified.
- (iii) The mobility of the carriers can be measured directly.
- (iv) It gives the concept of negative mass.

EXERCISES

THEORETICAL QUESTIONS

- 9.1** Explain Kronig-Penney model for the motion of electron in a periodic potential. What do you mean by negative mass ? Explain. [GGSIPU, May 2014 (8 marks)]
- 9.2** Explain Kronig-Penney model for the motion of an electron in a periodic potential.
- 9.3** What is the effect of periodic potential on the energy of electrons in a metal ?
- 9.4** Discuss the origin of energy bands in solids on the basis of Kronig-Penney model.
- 9.5** Discuss the Kronig-Penney model of an electron in a periodic potential. Show from $E-k$ diagram that materials can be classified into conductors, insulators and semiconductors. [GGSIPU, May 2014 reappear (6.5 marks)]
- 9.6** Discuss the origin of energy bands in solids. How can you distinguish between metals, semiconductors and insulators on the basis of energy bands ?
- 9.7** Discuss an electron in periodic field of a crystal i.e., the Kronig-Penney model, qualitatively.
- 9.8** Discuss the essential features of the electron energy band structure of solids on the basis of Kronig Penney model.
- 9.9** What are Brillouin zones ? Illustrate by constructing two Brillouin zones for a square lattice.
- 9.10** What is the physical meaning of Brillouin zones ? Explain.
- 9.11** What are Brillouin zones ?
- 9.12** What are Brillouin zones ? Sketch the first three Brillouin zones for a two dimensional square lattice.
- 9.13** Explain $E-k$ diagrams and Brillouin zones.
- 9.14** Discuss significance of $E-k$ curve.
- 9.15** Discuss the motion of electron in a periodic field of a crystal, now the effective mass of an electron in crystal is inversely proportional to second derivative of $E-k$ curve. Under what conditions the effective mass of an electron can become +ve, -ve and infinity ?
- 9.16** Discuss how the concepts of bands originated in solids. Give necessary theory. What are $E-k$ diagrams and what do you infer from them ?

- 9.17 What are $E-k$ diagrams ? What do you mean by negative mass ? Explain.
- 9.18 Discuss the origin of energy bands in solids on the basis of Kronig-Penney model.
[GGSIPU, May 2014 reappear (2.5 marks)]
- 9.19 Define the effective mass of electron.
- 9.20 Discuss the concept of effective mass of electrons in energy bands. Show that the effective number of electron in a filled band is zero.
- 9.21 Discuss effective mass of an electron and explain its physical significance.
- 9.22 Prove that for a completely filled, the number of effective electrons vanishes.
- 9.23 Calculate expression for effective mass of an electron moving in a periodic potential.
- 9.24 Based on band theory of solids, distinguish among conductor, semiconductor and insulator.
- 9.25 Prove that at absolute zero, all the states above Fermi level E_F are empty when temperature is raised above absolute zero.
- 9.26 Distinguish among metals, semiconductors and insulators on the basis of band theory and show that the effective number of free electrons in a completely filled band vanishes.
- 9.27 How do you explain the origin of energy bands in solids ? Explain with the support of a simple model.
- 9.28 How do energy bands originate ? Explain quantum mechanically.
- 9.29 Enumerate the difference between semiconductors and normal conductors in terms of their energy gap.
[GGSIPU, April 2005 (2 marks)]
- 9.30 Why do energy bands arise in solids ? Distinguish between conductors, semiconductors and insulators on the basis of band theory.
[GGSIPU, May 2007 (3 marks)]
- 9.31 Explain the origin of formation of energy bands. How does the band theory of solids lead to the classification of solid into conductors, semiconductors and insulators ?
[GGSIPU, May 2006 (3 marks)]
- 9.32 Compare the properties of metals, semiconductors and insulators on the basis of the energy band structure.
[GGSIPU, May 2010 (5.5 marks)]
- 9.33 Explain formation of energy bands in solids.
[GGSIPU, May 2010 (3 marks)]
- 9.34 Why do energy bands arise in solids ? Distinguish between conductors, semiconductors and insulators on the basis of band theory.
[GGSIPU, April 2011 (3 marks)]
- 9.35 Describe the behaviour and properties of conductors, insulators and semiconductors on the basis of band theory.
[GGSIPU, May 2014 (reappear) (3.5 marks)]
- 9.36 Describe the energy bands in conductors, semiconductors and insulators.
[GGSIPU, April 2011 (reappear) (3 marks)]
- 9.37 Describe the energy bands in conductors, semiconductors and insulators.
[GGSIPU, May 2012 (4-5 marks)]
- 9.38 Differentiate a conductor, a semiconductor and an insulator using energy-band diagram.
[GGSIPU, April 2013 (2 marks)]
- 9.39 Outline the formation of allowed and forbidden bands using the concept of periodicity of a crystal lattice.
[GGSIPU, April 2011 (4 marks)]
- 9.40 What are the Brillouin zones ? Draw the Brillouin zones for a one dimensional lattice of side ' a '.
[GGSIPU, May 2014 (2 marks)]
- 9.41 Name at least three semiconductors alongwith values of band gaps.

- 9.42 Distinguish between intrinsic and extrinsic semiconductors. Discuss the location of Fermi levels under suitable limiting conditions with necessary theory.
- 9.43 What is an intrinsic semiconductor? Obtain an expression for the electrical conductivity of an intrinsic semiconductor. Also derive the law of mass action for semiconductors.
- 9.44 Distinguish between intrinsic and extrinsic semiconductors.
- 9.45 Derive an expression for the electron (or carrier) concentration in an intrinsic semiconductor.
- 9.46 Derive an expression for the concentration (or density) of holes in the valence band of an intrinsic semiconductor.
- 9.47 Derive an expression for the density of electrons (or electron concentration) in the conduction band of an N-type semiconductor. What happens to the Fermi level as temperature increases?
- 9.48 Derive an expression for the density of holes in the valence band of a P-type semiconductor. Show the variation in position of Fermi level.
- 9.49 Why does Fermi level lie near conduction band in N-type and near valence band in P-type semiconductor.
- 9.50 What happens to Fermi level as a result of doping?
- 9.51 Show that the product of concentration of electrons in the conduction band and that of holes in the valence band of a semiconductor is independent of Fermi level but depends upon the energy gap. Hence find their separate values in terms of energy gap.
- 9.52 What is Fermi energy? What is its physical significance? [IGGSIPU, May 2008 (6.25 marks)]
- 9.53 What is the significance of Fermi energy in a fermion system at 0 K; At $T > 0$ K? [IGGSIPU, April 2008 (3 marks)]
- 9.54 Draw the Fermi energy level diagram for an intrinsic and extrinsic semiconductors. [IGGSIPU, April 2009[Reappear] (3 marks)]
- 9.55 Where does Fermi level lie in an N-type semiconductor? Explain with schematic diagram. [IGGSIPU, May 2010 (2 marks)]
- 9.56 Discuss the effect of temperature and doping concentration on the position of Fermi level in a N-type semiconductor. [IGGSIPU, April 2008 (2 marks)]
- 9.57 Write short notes on
 (i) Fermi level and Fermi energy.
 (ii) Energy level diagram for intrinsic and doped semiconductors. [IGGSIPU, April 2007 (2.5 marks)]
- 9.58 Mark the Fermi level for
 (i) an intrinsic semiconductor
 (ii) N-type semiconductor and
 (iii) P-type semiconductor. [IGGSIPU, May 2014 reappear (3 marks)]
- 9.59 Explain why a semiconductor acts as an insulator at 0 K. [IGGSIPU, April 2008 (3 marks)]
- 9.60 Discuss the role of Fermi energy level in intrinsic semiconductors. How it is affected by doping? [IGGSIPU, May 2009 (4 marks)]
- 9.61 How does Fermi energy change with temperature in intrinsic semiconductor? [IGGSIPU, May 2014 reappear (2.5 marks)]

- 9.62 Explain the effect of temperature on the Fermi level of N-type of semiconductor. [GGSIPU, May 2014 (2 marks)]
- 9.63 What is Hall effect? Explain briefly.
- 9.64 What is Hall effect? Give an elementary theory of Hall effect. Mention applications of Hall effect.
- 9.65 What is Hall effect? Give an experimental method of calculating concentration and type of charges in a given semiconductor.
- 9.66 Show that the Hall-coefficient is independent of applied magnetic field and is inversely proportional to the current density and electronic charge.
- 9.67 What is Hall effect? Define Hall coefficient. List its applications.
- 9.68 Define Hall coefficient and establish its relation with the density of charge carriers.
- 9.69 Describe a method which can be used to measure the Hall coefficient of solids with one type of charge carriers.
- 9.70 What is Hall effect? Derive an expression for Hall coefficient. Discuss some of its important applications. [GGSIPU, May 2014 (6 marks)]
- 9.71 Define 'Hall effect' and 'Hall coefficient'.
- 9.72 Derive an expression for Hall coefficient.
- 9.73 Write notes on the following : (i) Hall effect (ii) Effective mass.
- 9.74 Define 'effective mass'. How is 'effective mass' of an electron different from the free electron mass?
- 9.75 Define Hall effect and Hall coefficient. Briefly discuss the physical origin of Hall effect. What are uses of this effect? [GGSIPU, May 2014 (reappear) (6 marks)]

NUMERICAL PROBLEMS

- 9.1 Consider two dimensional square lattice of side $3 \text{ \AA}$. At what electron momentum value do the sides of first Brillouin zone appear? What is the energy of free electron with this momentum?

Hint. $p = \hbar k$

for FBZ $k = \pm \pi / a$

$$p = \frac{\hbar \pi}{a} = \frac{\hbar}{2\pi} \times \frac{\pi}{a} = \frac{\hbar}{2a} = 1.1 \times 10^{-24} \text{ kg m/sec}$$

Energy $E = \frac{p^2}{2m} = 6.688 \times 10^{-19} \text{ J}$

- 9.2 The energy of an electron in a crystalline solid is related to the wave vector k by the relation

$$E = 10 \frac{\hbar^2 k^2}{m}$$

Calculate the velocity and effective mass.

Hint.

$$v = \frac{1}{\hbar} \frac{dE}{dk} = \frac{1}{\hbar} \frac{d}{dk} \left(\frac{10 \hbar^2 k^2}{m} \right) = \frac{20 \hbar m}{m}$$

and

$$m^* = \frac{\hbar^2}{\left(\frac{d^2 E}{dk^2} \right)} = \frac{\hbar^2 m}{20 \hbar^2} = \frac{m}{20}$$

- 9.3 The energy gap of two intrinsic semiconductors A and B are 0.36 eV and 0.72 eV respectively. Compare the intrinsic carrier density of A to B at 300 K. (Given $m_h^* = m_e^* = 9.1 \times 10^{-31}$ kg and $2k_B T = 0.052$ eV).

Hint.
$$n_i = 2 \frac{2\pi k_B T m}{h^2} \times \frac{m_e^* m_h^*}{m^2} \exp\left(-\frac{E_g}{2k_B T}\right)$$

Then
$$\frac{(n_i)_A}{(n_i)_B} = \exp\left[\frac{0.36 \text{ eV}}{0.052 \text{ eV}}\right] = 1.0154 \times 10^3.$$

The ratio of the intrinsic carrier densities of the materials A and B is 1.0154×10^3 .

- 9.4 Show that the integral

$$I = \int_{E_C}^{\infty} (E - E_C)^{1/2} e^{-E/k_B T} dE = (k_B T)^{3/2} e^{-\frac{E_C}{k_B T}} \sqrt{\frac{\pi}{2}}$$

Hint. $\frac{E - E_C}{k_B T} = x$; so that $(E - E_C) = k_B T x$;

$$\frac{E}{k_B T} = \frac{E_C}{k_B T} + x \text{ and } dE = k_B T dx$$

$$I = (k_B T)^{3/2} e^{-E_C/k_B T} \int_0^{\infty} x^{1/2} e^{-x} dx = (k_B T)^{3/2} e^{-E_C/k_B T} \frac{\sqrt{\pi}}{2}$$

- 9.5 In a n -type semiconductor, the Fermi level lies 0.3 eV below the conduction band at 300 K. If the temperature is increased to 330 K. Find the new position of Fermi level.

Hint. At 300 K E_F lies 0.3 eV below E_C

We know that
$$E_C - E_F = k_B T \ln \frac{N_c}{N_d}$$

$$\Rightarrow 0.3 \text{ eV} = 300 k_B T \ln \frac{N_c}{N_d}$$

At 330 K
$$E_C - E_F = 330 k_B T \ln \frac{N_c}{N_d}$$

Dividing Eq. (ii) by Eq. (i), we get
$$\frac{E_C - E_F}{0.3} = \frac{330}{300} \text{ or } E_C - E_F = 0.3 \text{ eV}$$

Thus the Fermi level lies 0.33 eV below the conduction band.

- 9.6 Fermi energy of an intrinsic semiconductor is 0.6 eV. The low lying energy levels in conduction band is 0.2 eV above the Fermi level. Calculate the probability of occupation of this level by an electron at room temperature.

Hint. $E_F = 0.6 \text{ eV}$, $E = (0.6 + 0.2) \text{ eV} = 0.8 \text{ eV}$

Probability $f(E) = \frac{1}{1 + \exp \frac{E - E_F}{k_B T}} = 0.0004 = 0.04\%$

- 9.7 A germanium N-type semiconductor has donor density of $10^{21} / \text{m}^3$. Find Hall voltage when a magnetic field of 0.5 Wb/m^2 is applied and current density = 500 A/m^2 . The thickness of the semiconductor = 3 mm .

Hint. $V = \frac{E_y}{d} = \frac{J_x B_z d}{E_y} = \frac{500 \times 0.5 \times 3 \times 10^{-3}}{10^{21} \times 1.6 \times 10^{-19}} = 4.68 \times 10^{-3} \text{ V} = 4.68 \text{ mV}$

- 9.8 An N-type germanium sample has donor density of $10^{21} / \text{m}^3$. It has arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m^2 . Find the Hall voltage if the sample is 3 mm wide.

Hint. $R_H = \frac{1}{Ne} = \frac{E_y}{J_x B_z} = \frac{V_H}{t J_x B_z} = \frac{(3 \times 10^{-3}) \times 500 \times 0.5}{10^{21} \times (1.6 \times 10^{-19})} \text{ V} = 4.69 \times 10^{-3} \text{ V} = 4.69 \text{ mV}$

- 9.9 When 90 mA current is passed through a sodium specimen under the magnetic field 2.0 Wb/m^2 , the Hall voltage is 0.09 mV . The width of the specimen is 0.04 mm . Calculate carrier concentration.

Hint. $N = \frac{t I_x B_z}{V_H t e} = \frac{90 \times 10^{-3} \times 2.0}{0.09 \times 10^{-3} \times 0.04 \times 10^{-3} \times 1.6 \times 10^{-19}} = 3.1 \times 10^{26} / \text{m}^3$

- 9.10 Assuming there are 5×10^{28} atoms/ m^3 in copper, find Hall coefficient.

Hint. $R_H = \frac{-1}{ne} = -0.125 \times 10^{-9} \text{ m}^3 / \text{C}$

- 9.11 Hall voltage of 1.0 mV to be developed when a sample carrying a current of 10.0 mA is placed in a transverse magnetic field of 0.3 T . Calculate the charge carrier concentration of the sample, given thickness of sample along the direction of magnetic field is 0.3 mm .

Hint. $n = \frac{I B_z}{V_H b e} = 6.25 \times 10^{22} \text{ m}^{-3}$

- 9.12 Calculate the value of Hall voltage is 30 mV for a current of 70 mA flowing through a specimen of 3 mm thick placed in a magnetic field of strength 6×10^3 Gauss. ($1 \text{ Gauss} = 10^{-4} \text{ Tesla}$)

Hint. $R_H = \frac{b V_H}{I B_z} = 2.14 \times 10^{-3} \text{ m}^3 / \text{C}$

and $R_H = \frac{1}{n}$

or $n = \frac{1}{e R_H} = 2.91 \times 10^{21} \text{ m}^{-3}$

- 9.13 The Hall coefficient of certain silicon specimen was found to $-7.35 \times 10^{-5} \text{ m}^3 \text{C}^{-1}$ from 100 to 400 K . Determine the nature of semiconductor. If the conductivity was found to be $200 \text{ m}^{-1} \Omega^{-1}$, calculate the density and mobility of charge carrier. [GGSIPU, May 2014 (4.5 marks)]

Hint. The negative sign of the Hall coefficient indicates that the nature of semiconductor is N-type.

$$R_H = -\frac{1}{ne} \quad \text{or} \quad n = \frac{1}{e R_H} = \frac{1}{(1.6 \times 10^{-19}) \times (7.35 \times 10^{-5})} = 8.503 \times 10^{22} \text{ m}^{-3}$$

and $\sigma = ne\mu \Rightarrow \mu = \frac{\sigma}{ne} = \frac{200}{(8.503 \times 10^{22})(1.6 \times 10^{-19})} = 1.47 \times 10^{-2} \text{ m}^2 \text{V}^{-1} \text{s}^{-1}$

- 9.1 The energy near the valence band edge of a crystal is given by $E = -10^{-39} k^2 \text{ Jm}^2$. An electron with wave vector $10^{10} \hat{k}_x \text{ m}^{-1}$ is removed from an orbital in the completely filled valence band. Determine effective mass and momentum. [Ans. $m^* = -5.5 \times 10^{-30} \text{ kg}$, $p = \hbar k = 1.05 \times 10^{-24} \hat{k}_x \text{ Js/m}$]
- 9.2 In an N-type semiconductor the Fermi level lies 0.3 eV below the conduction band at 300 K. If the temperature is increased to 330 K, find the new position of the Fermi level. [Ans. 0.33 eV below the conduction band]
- 9.3 Calculate the Fermi level position in Si doped with 10^{15} phosphorus atoms/ m^3 at room temperature assuming complete ionization. From the calculated Fermi level position, check the assumption of complete ionization of carriers. Given that $N_C = 2.8 \times 10^{19} / \text{cm}^3$ and $(E_C - E_d) = 0.148 \text{ eV}$. [Ans. 99% of the donor atoms are ionized]
- 9.4 In a P-type semiconductor, the Fermi level lies 0.4 eV above the valence band. If the concentration of the acceptor atom is tripled find new position of the Fermi level. [Ans. $(E'_F - E_V) = 0.367 \text{ eV}$]
- 9.5 When two metals are joined together, their Fermi levels must be equal, so there is a transfer of electrons from the metal of higher Fermi level to one of the lower ones, until the contact potential is equal to the original difference in Fermi level. Estimate the contact potential for copper and gold. The Fermi energy of copper is 7 eV and for gold is 5.5 eV. [Ans. $E_F - E'_F = 1.5 \text{ eV}$]
- 9.6 Calculate the position of Fermi level E_F at 300 K for germanium crystal containing 5×10^{22} arsenic atom/ m^3 . [Ans. $E_C - E_F = 0.16 \text{ eV}$]
- 9.7 5 μA current flows through a simple P-N junction diode at room temperature when it is reverse biased with 0.15 V. Determine the amount of current which will flow when it is forward biased with the same voltage. [Ans. 1.66 mA]
- 9.8 Find the fraction of electrons excited into conduction band in germanium at 100K and 300K.
- 9.9 In a P-type semiconductor, the Fermi level lies 0.4 eV above the valence band. If the concentration of the acceptor atom is tripled, find the new position of the Fermi level. [Ans. 0.367 eV]
- 9.10 Calculate the Hall voltage across the width of a semiconducting specimen from the following data:
width of the specimen = 0.1 m ; thickness of the specimen = 0.01 m ;
field applied perpendicular to both width and length = 0.6 T ;
current flowing length wise = 10 mA ; Hall coefficient = $3.84 \times 10^{-4} \text{ m}^3 / \text{C}$. [Ans. $2.3 \times 10^{-4} \text{ V}$]
- 9.11 Calculate the density of conduction electrons in Na, if its lattice parameter $a = 4.3 \text{ \AA}$. Also calculate the Hall coefficient. [Ans. $2.52 \times 10^{28} / \text{m}^3$; $-248 \times 10^{-10} \text{ m}^3 / \text{C}$]
- 9.12 Consider a slab of Cu, 0.1 mm thick, 1.0 mm wide and 10.0 mm long :
(a) If we drive a current of 1 A down the length of slab, what is the current density ?
(b) If we then put the slab in the magnetic field of 1T with the field perpendicular to the $1 \text{ mm} \times 10 \text{ mm}$ face, what is Hall field will be produced, if the Hall coefficient is $0.55 \times 10^{-10} \text{ m}^3 / \text{C}$.
(c) What Hall voltage will we observe across the slab ?
[Ans. (a) 10^7 A/m^2 ; (b) $5.5 \times 10^{-4} \text{ V/m}$; (c) $0.55 \mu\text{V}$]